

Assembly and User Guide

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1

Introduction to eArm Robot

The eArm robot is an ESP32-powered robotic arm. It offers a hands-on way to explore robotics, coding, and electronics in an engaging platform.

Features:

① Quick Assembly & Ready to Use

► The eArm robot arm is designed for effortless assembly—you can build it and start experimenting within 1-2 hours.

② Pre-Programmed for Instant Operation

► The control board comes pre-flashed with firmware, eliminating the need for initial code uploads. Simply power it on and begin using it right away.

③ Integrated Power Management

► Features an onboard battery holder with a power indicator. The robot arm can be charged directly via USB, ensuring uninterrupted operation.

④ Precision Control with Joystick

► The included ergonomic joystick delivers stable control, offering intuitive and responsive manipulation of the robot arm.

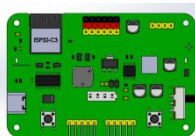
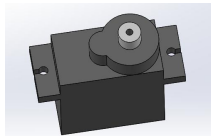
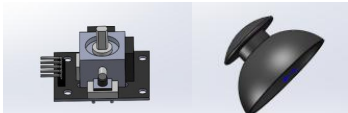
⑤ Web-Based Control

► Supports cross-platform control via a dedicated Web App, accessible from any device with a browser—no additional installations required.

⑥ Expandable Learning with Arduino

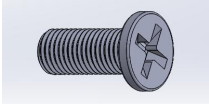
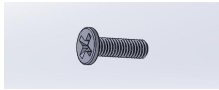
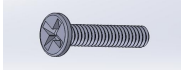
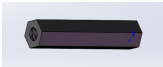
► Includes step-by-step Arduino programming tutorials, helping users advance from basic operations to custom robotics development.

1.1 Package Contents

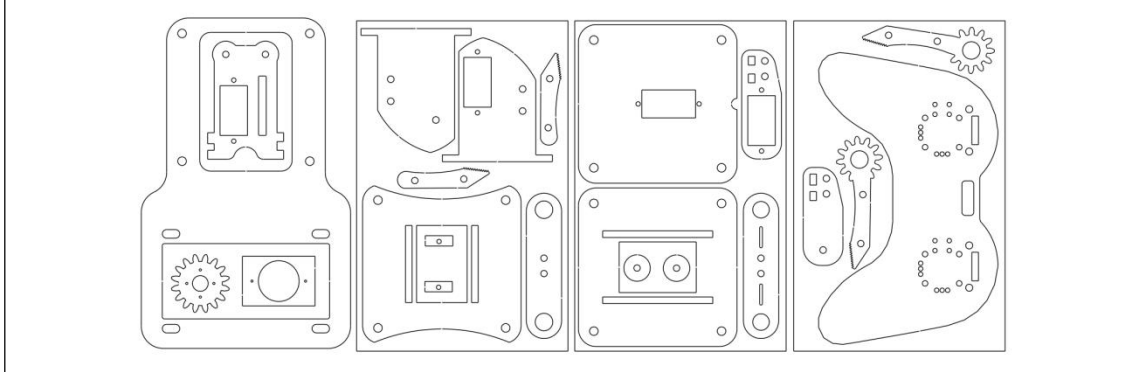
#Part 1 : Electronics Components		
Control Board 1PCS	Servo 4 PCS	5P Dupont Wire 2PCS
		
Joystick 2PCS	USB Cable 1PCS	
		

#Part 2 : Fasteners / Accessories

The following accessories listed in sequential order are all packaged in a small bag with a label and serial number attached. You can quickly locate them by referring to the serial number and name on the bag.

1 M4×10mm Screw 21Pcs	2 M3×14mm Screw 3Pcs	3 M3×10mm Screw 19Pcs
		
4 M2×10mm Screw 9Pcs	5 M1.4×5mm Screw 8Pcs	6 M4 Nut 5Pcs
		
7 M3 Lock Nut 9Pcs	8 M2 Nut 10Pcs	9 M3×7mm Standoff 5Pcs
		
10 M3×28mm Standoff 4Pcs	11 M3×40mm Standoff 2Pcs	12 M4×20mm Standoff 8Pcs
		
13 M3×6mm Screw 9Pcs	14 Flanged Bearing 4Pcs	Rotating Base 1PCS
		

#Part 3 : Acrylic Sheet



#Part 4 : Tools

		
M3 screwdriver 1PCS	M1.5 screwdriver 1PCS	wrench 1PCS



Note: You need to buy a 18650 lithium battery yourself!

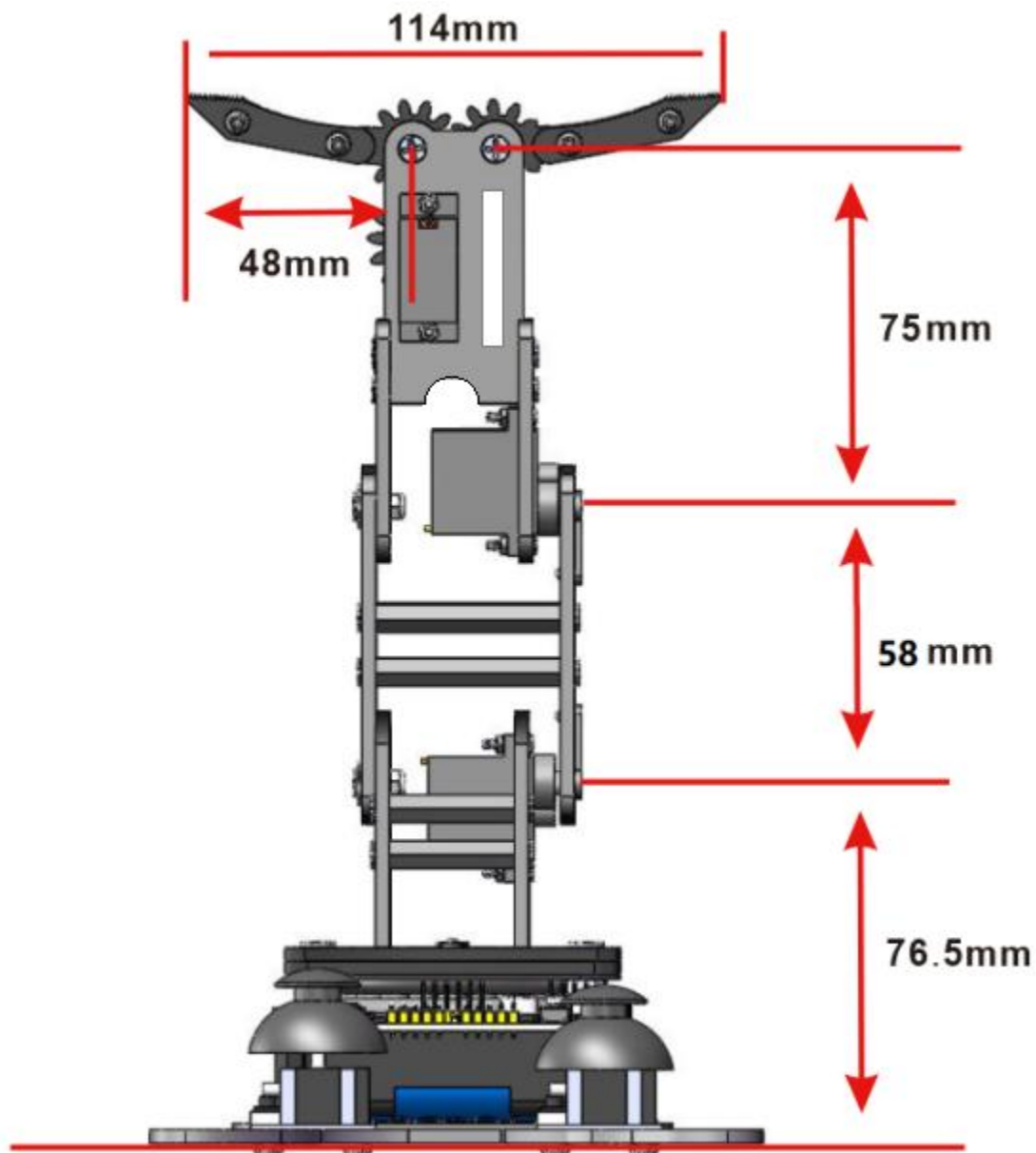
Parameters of 18650 lithium battery



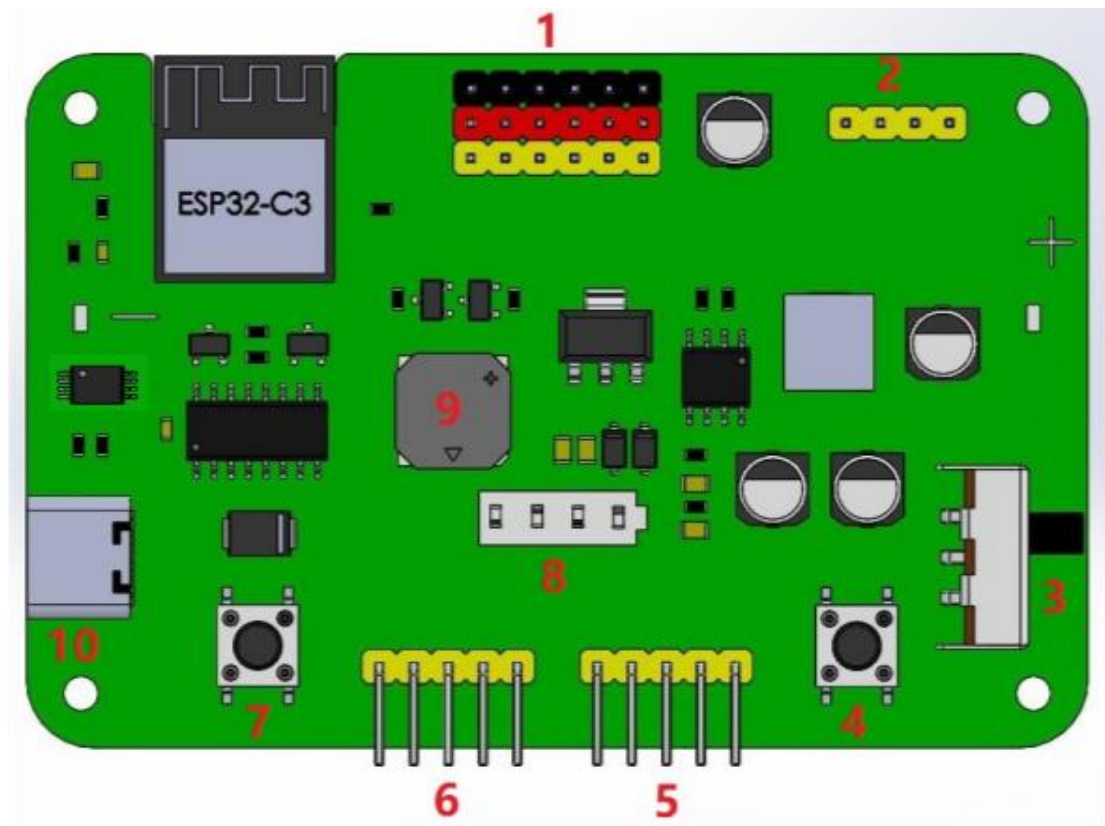
- Model: 18650 lithium battery
- Top type: Both button top and flat top are acceptable
- Capacity: >2000mAh
- Maximum charging voltage: 4.2V
- Nominal voltage: 3.7V
- End-off voltage: 2.75V
- Minimum charging current: >2A
- Minimum discharge current: >4A

1.2 Specifications:

Control Board	eArm-ESP32-C3-MINI-1
Programming language	Arduino\MicroPython
Recommended maximum lifting capacity:	40g
Charge/discharge	5V/2A
Servo type	MG90S 180°
Remote control method	Joystick or Web APP
Assembly time required	2-4 hours
Product size	15X9.5X26CM
Packaging size	18.5X12.5X4.3CM
Applicable age	experienced players of any age or beginner aged 12+
Battery required	a 18650 battery(Not included)



1.3 Introduction to ESP32 Control Board:



- 1: IO port, 5V/2A driving capability.
- 2: IO port and serial port expansion port.
- 3: Power switch.
- 4: Battery level display button(When the power switch is off, pressing it briefly can display the battery level).
- 5: IO port, 3.3V/500mA driving capability.
- 6: IO port, 3.3V/500mA driving capability.
- 7: Reset button.
- 8: Battery level indicator LED.
- 9: Buzzer.
- 10: USB Type-c Port for charging or uploading code

2

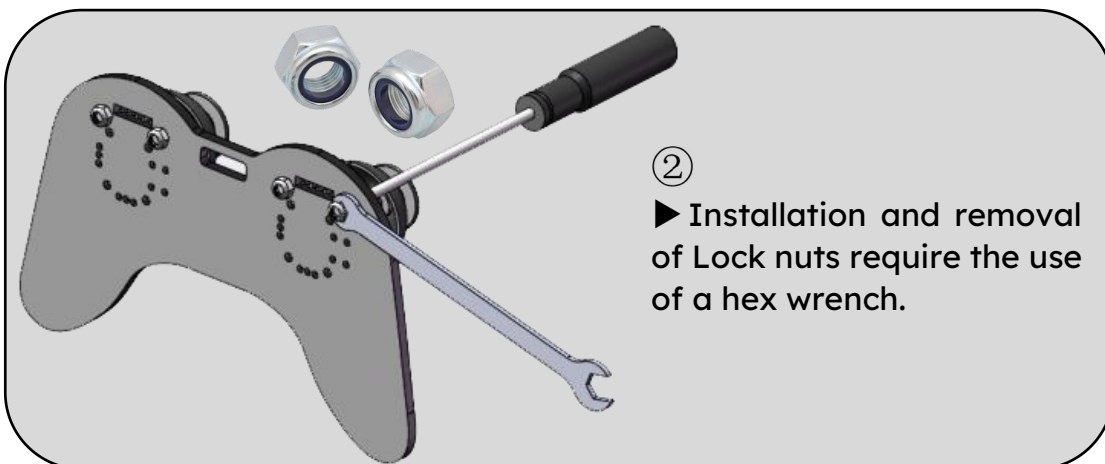
Assemble the eArm Robot

Before You Begin: Important Notes



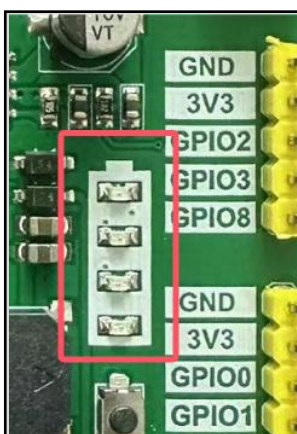
①

► The surface of the acrylic board is covered with a protective film, please tear it off before installation!



②

► Installation and removal of Lock nuts require the use of a hex wrench.



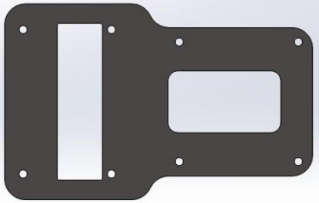
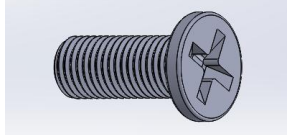
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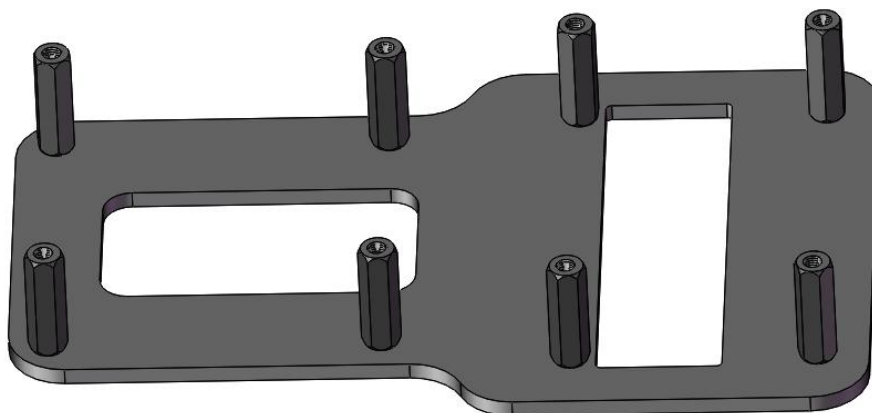
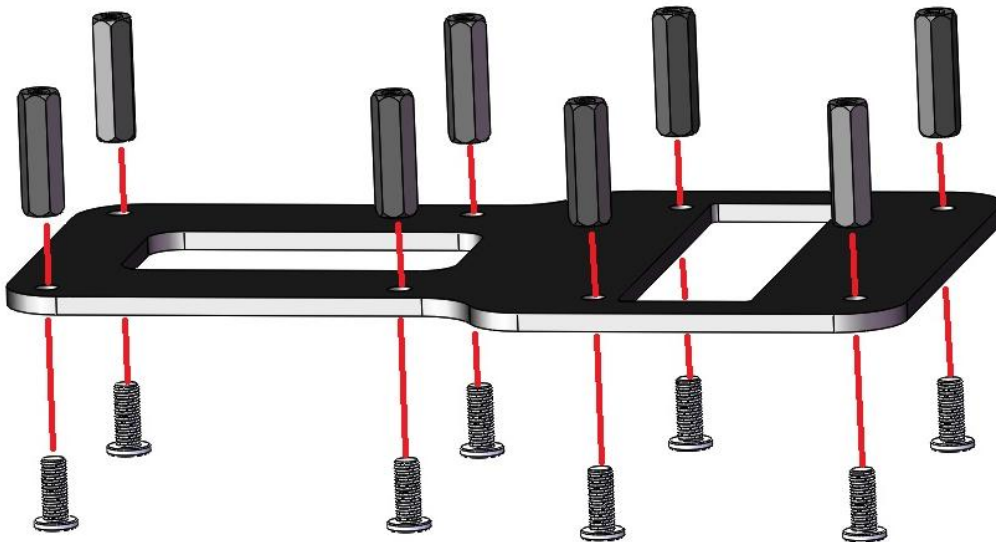


A minimum of three battery bars is required to ensure proper installation and operation of the robot. After installing an 18650 battery into the battery holder, check the power indicator. If it shows less than three bars, please charge the battery either:

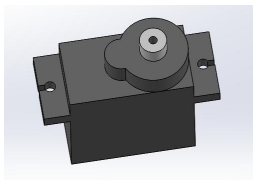
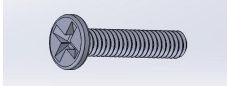
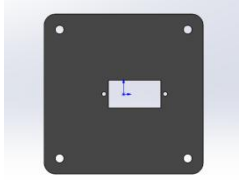
- Using an 18650 dedicated charger
- Via the board's charging port with a USB cable.

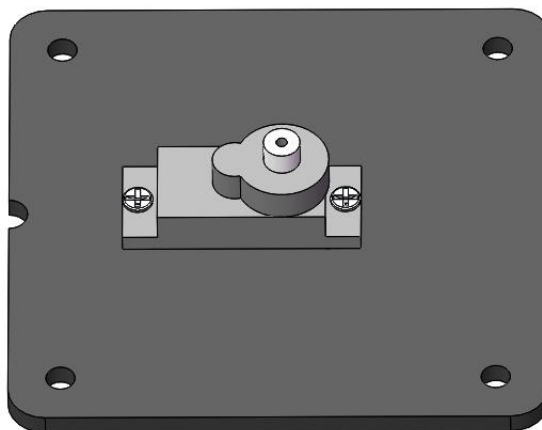
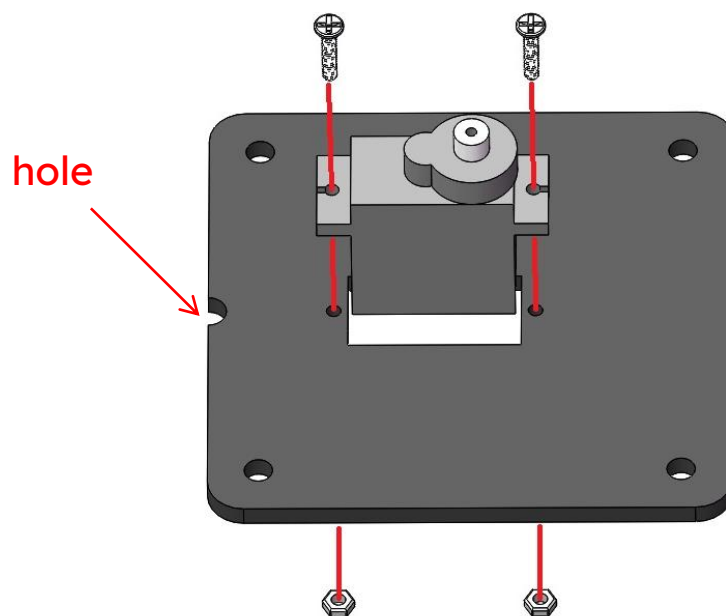
Step 1 Assemble the Robot Base

Acrylic Sheet 1PCS	Bag No.⑫ M4X20MM Standoff 8PCS	Bag No.① M4X10MM Screw 8PCS
		

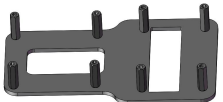
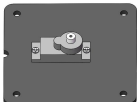
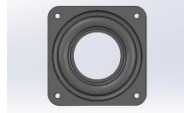
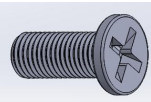


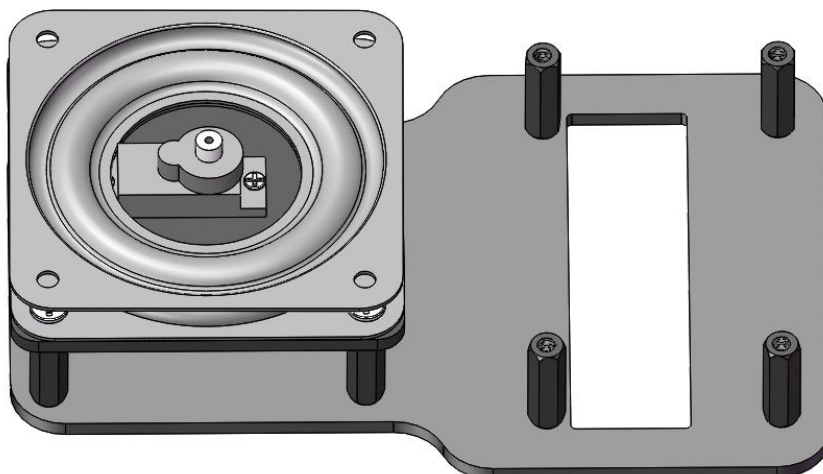
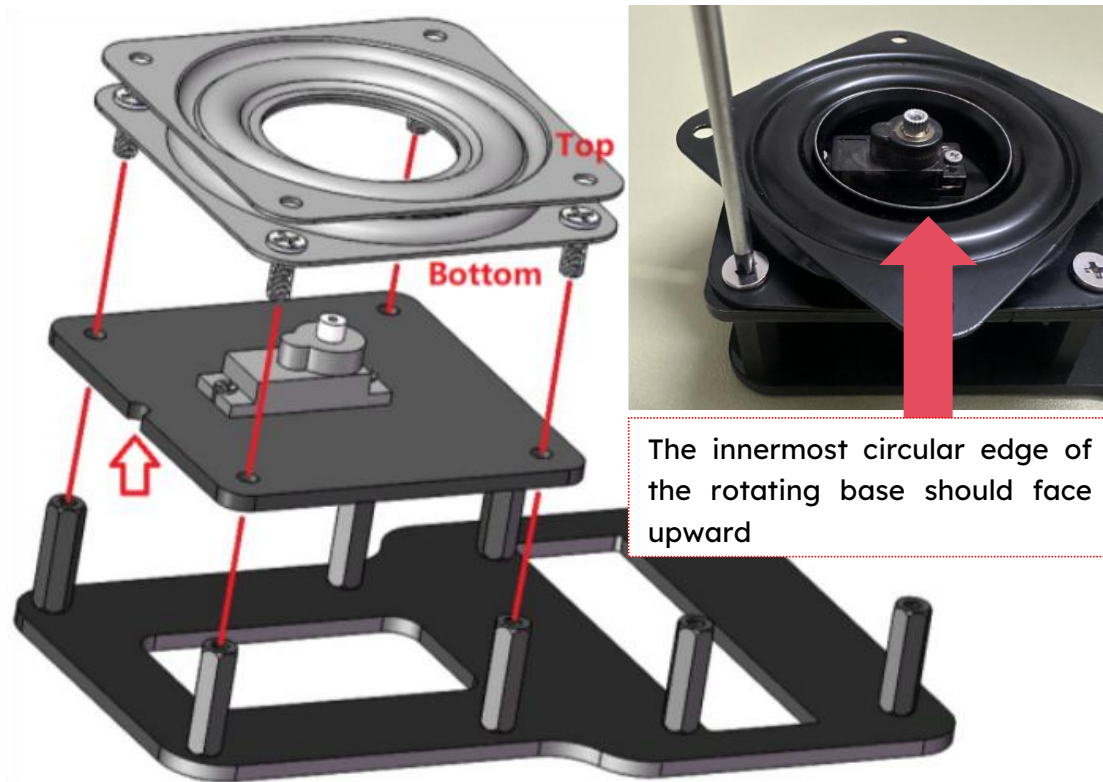
Step 2 Assemble Servo A on Robot Base

Servo 1 PCS	Bag No.④ M2x10MM Screw 2PCS	Bag No.⑧ M2 Nut 2PCS	Acrylic Sheet 1PCS
			

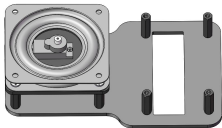
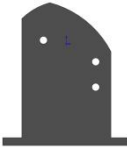
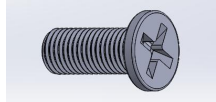
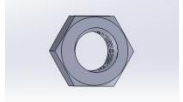


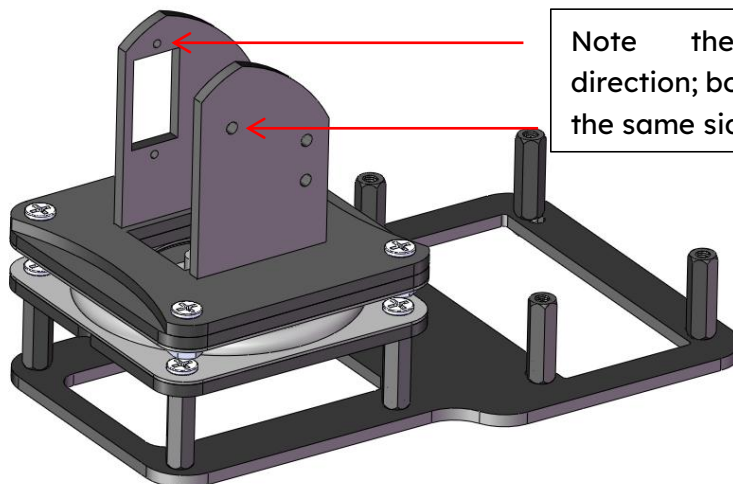
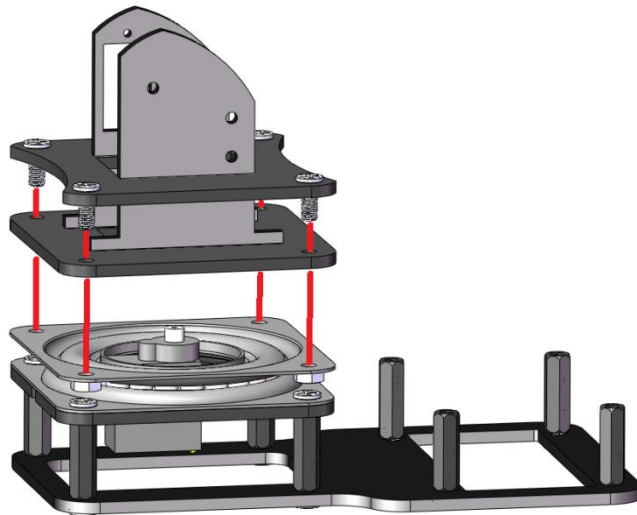
Step 3 Assemble the Rotating Base

Step 1 Structure	Step 2 Structure	Rotating Base 1PCS	Bag No.① M4X10MM Screw 4PCS
			



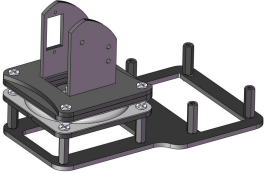
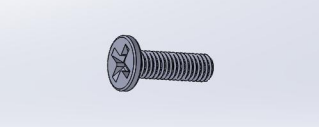
Step 4 Assemble the Lower arm

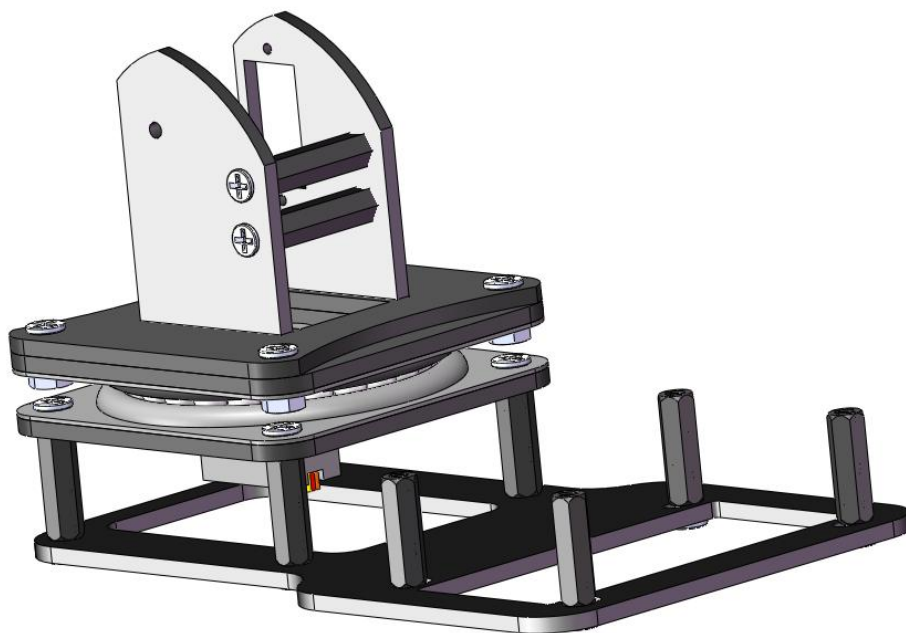
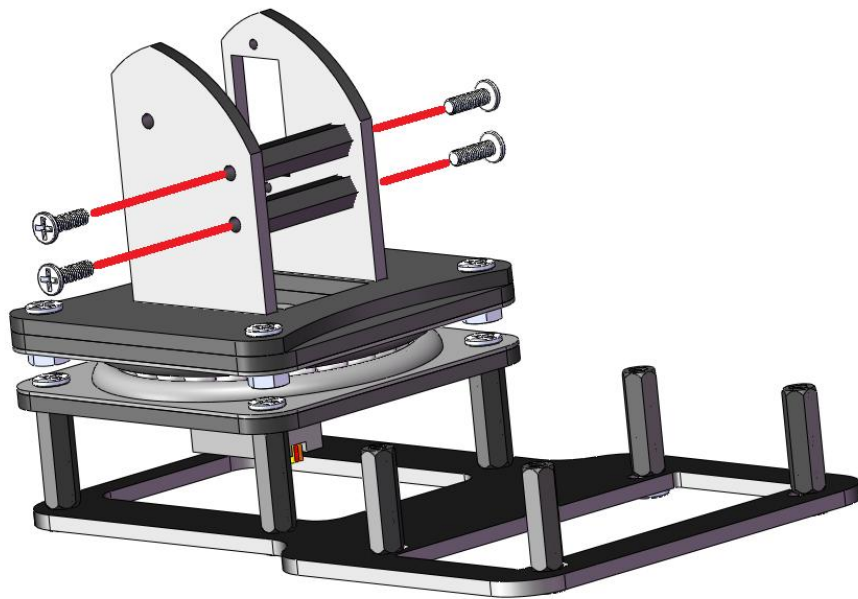
Step 3 Structure	Acrylic Sheet 1PCS		Acrylic Sheet 1PCS	
				
Acrylic Sheet 1PCS	Acrylic Sheet 1PCS	Bag No.① M4X10MM Screw 4PCS		Bag No.⑥ M4 Nut 4PCS
				



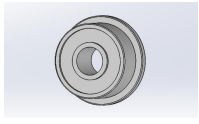
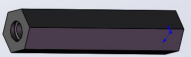
Note the installation direction; both holes are on the same side.

Step 5 Fix the lower arm

Step 4 Structure	Bag No.⑩ M3x28MM Standoff 2PCS	Bag No.③ M3X10MM Screw 4PCS
		

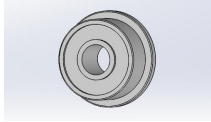
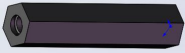


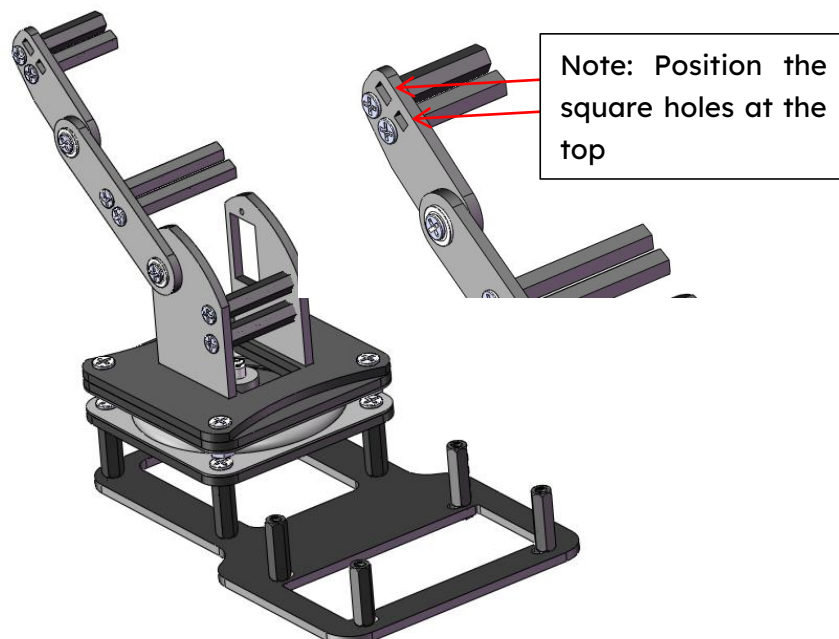
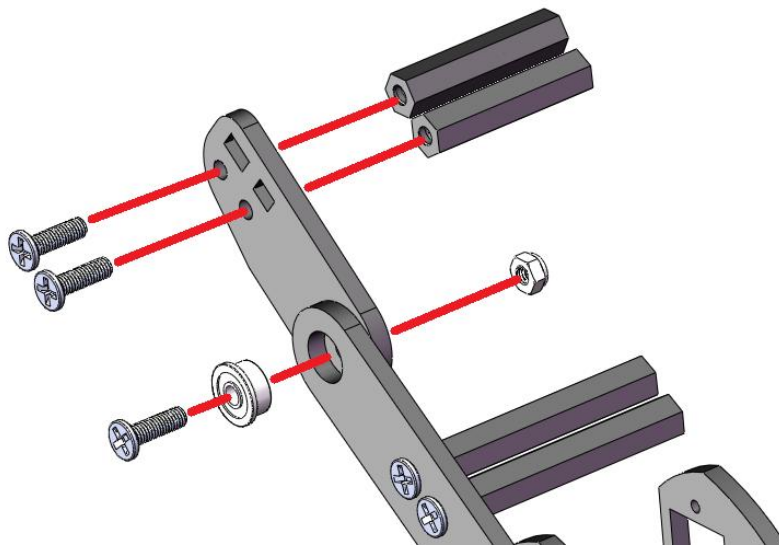
Step 6 Assemble the Middle Arm

Step 5 Structure	Acrylic Sheet 1PCS	Bag No.⑭ Flange Bearing 1PCS
		
Bag No.⑪ M3x40MM Standoff 2PCS	Bag No.③ M3X10MM Screw 3PCS	Bag No.⑦ M3 Lock Nut 1PCS
		

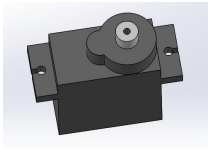
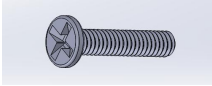
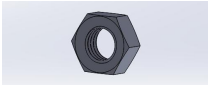


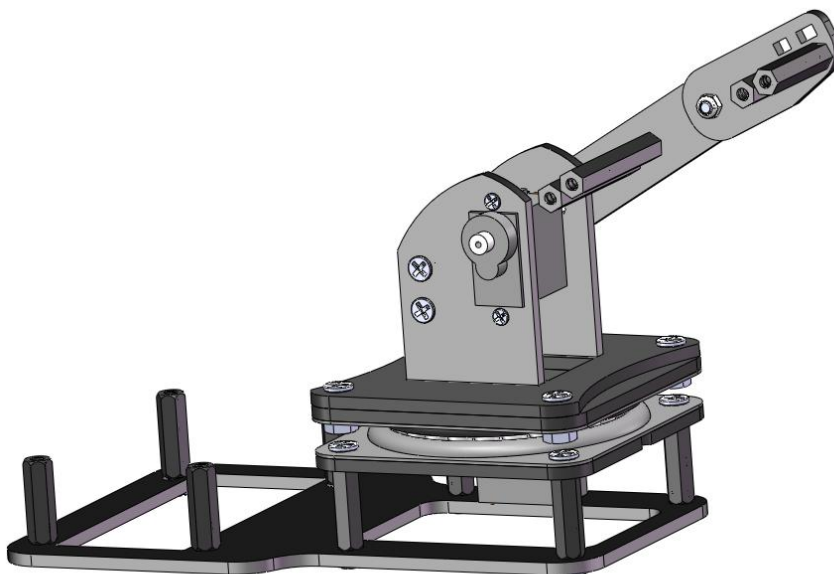
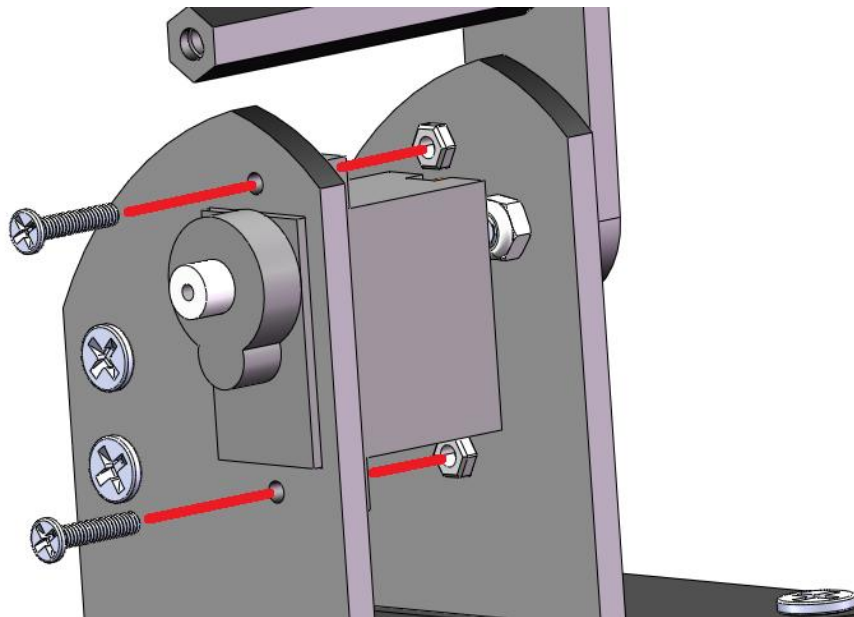
Step 7 Assemble the Upper Arm

Step 6 Structure	Acrylic Sheet 1PCS	Bag No.⑭ Flange Bearing 1PCS
		
Bag No.⑩ M3x28MM Standoff 2PCS	Bag No.③ M3X10MM Screw 3PCS	Bag No.⑦ M3 Lock Nut 1PCS
		

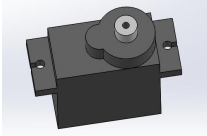
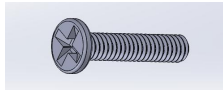
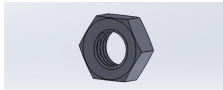


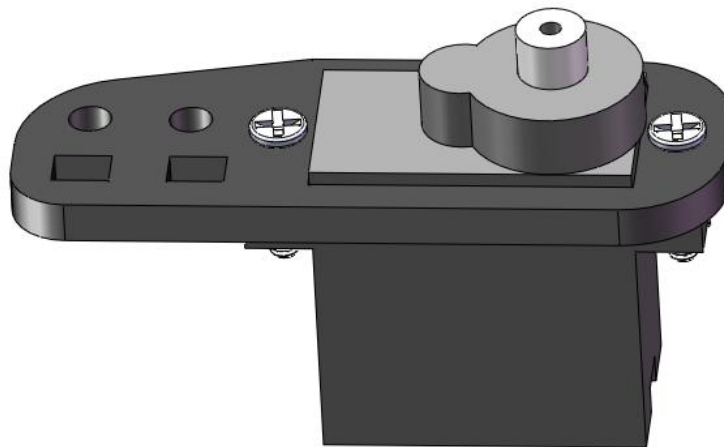
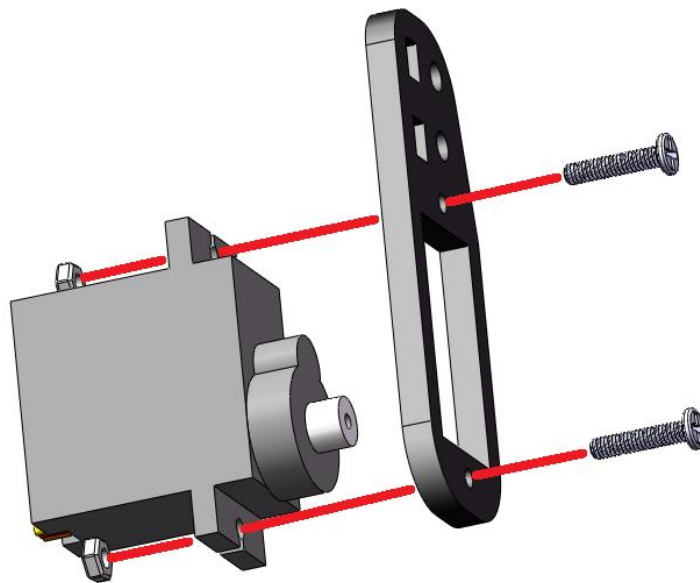
Step 8 Assemble the Servo B on lower arm

Step 7 Structure	Servo 1PCS	Bag No.④ M2x10MM Screw 2PCS	Bag No.⑧ M2 Nut 2PCS
			

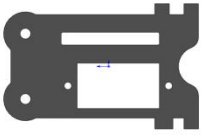
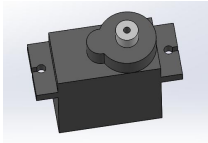
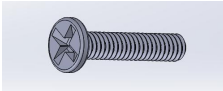
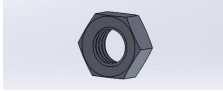


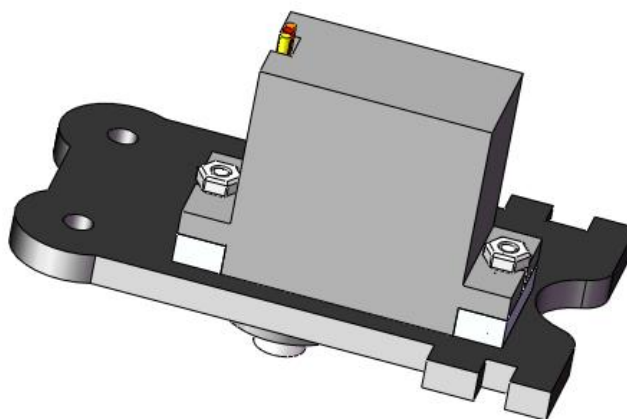
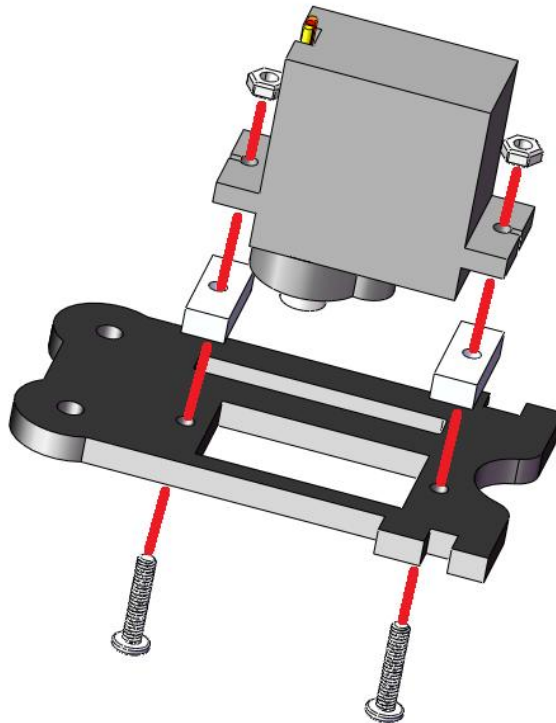
Step 9 Assemble the Servo C on Middle Arm

Acrylic Sheet 1PCS	Servo 1PCS	Bag No.④ M2x10MM Screw 2PCS	Bag No.⑧ M2 Nut 2PCS
			

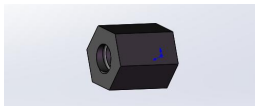
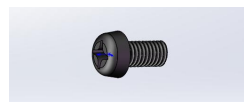


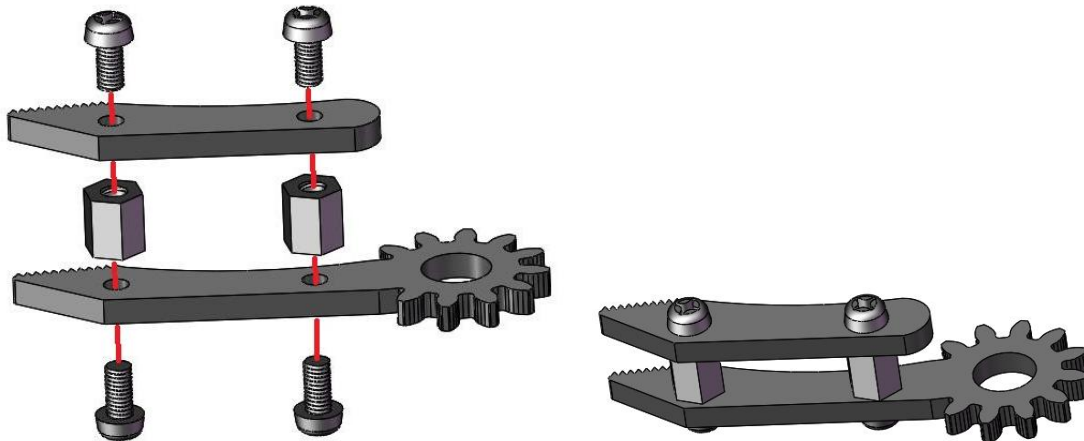
Step 10 Assemble the Servo D on Upper Arm

Acrylic Sheet 1PCS		Servo 1PCS	
			
Acrylic Sheet 2PCS	Bag No.④ M2x10MM Screw 2PCS	Bag No.⑧ M2 Nut 2PCS	
			

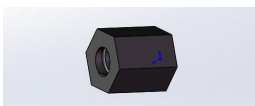
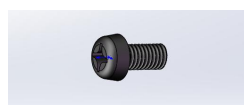


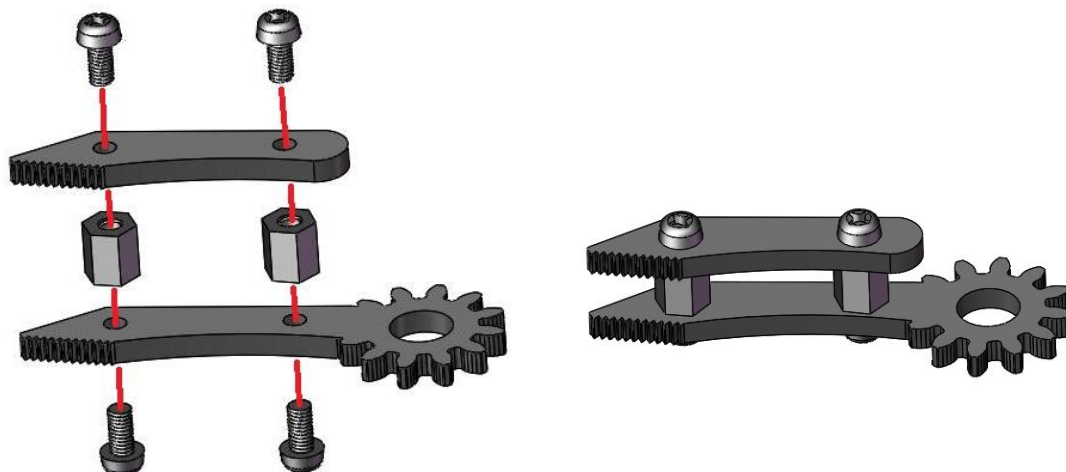
Step 11 Assemble the Left Gripper Plate of The Clamp

Acrylic Sheet 1PCS	Acrylic Sheet 1PCS	Bag No.⑨ M3x7MM Standoff 2PCS	Bag No.⑬ M3x6MM Screw 4PCS
			

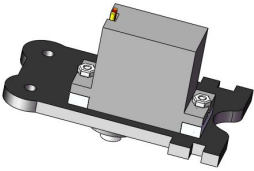
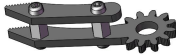
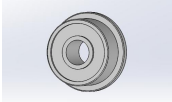


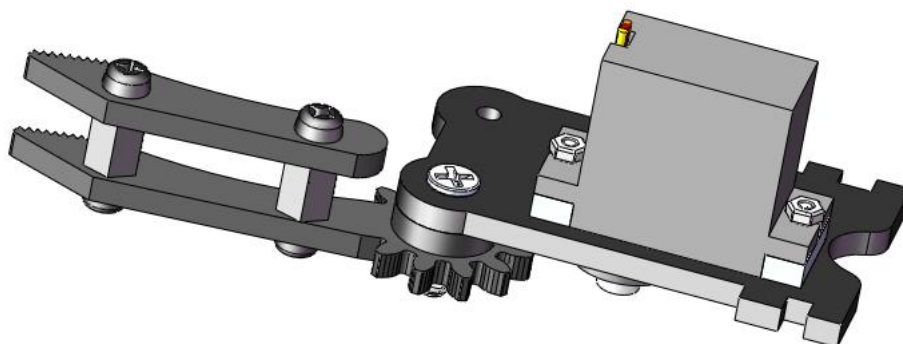
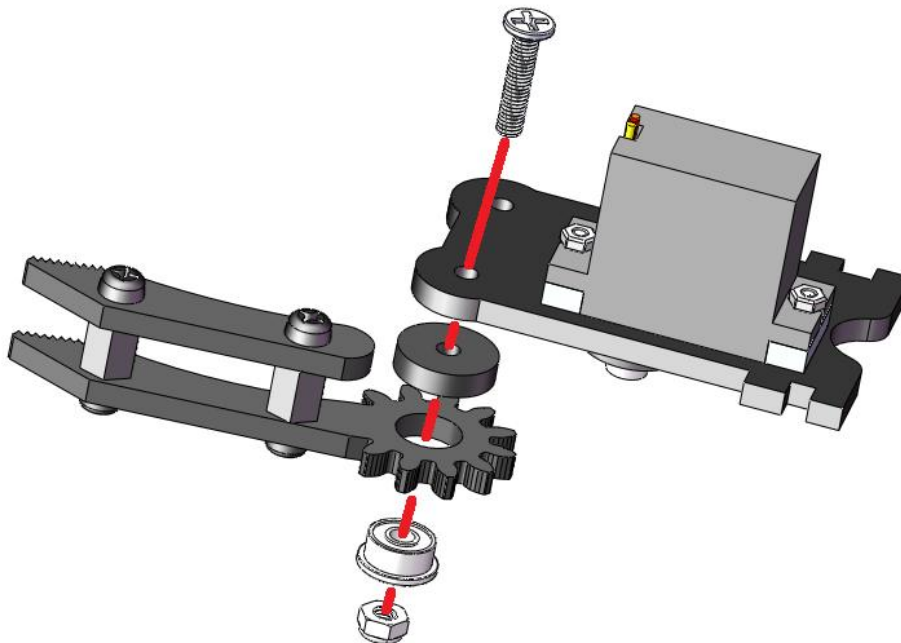
Step 12 Assemble the Right Gripper Plate of The Clamp

Acrylic Sheet 1PCS	Acrylic Sheet 1PCS	Bag No.⑨ M3x7MM Nylon Column 2PCS	Bag No.⑬ M3x6MM Nylon Screw 4PCS
			

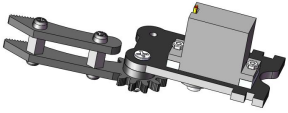
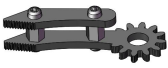
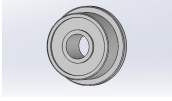


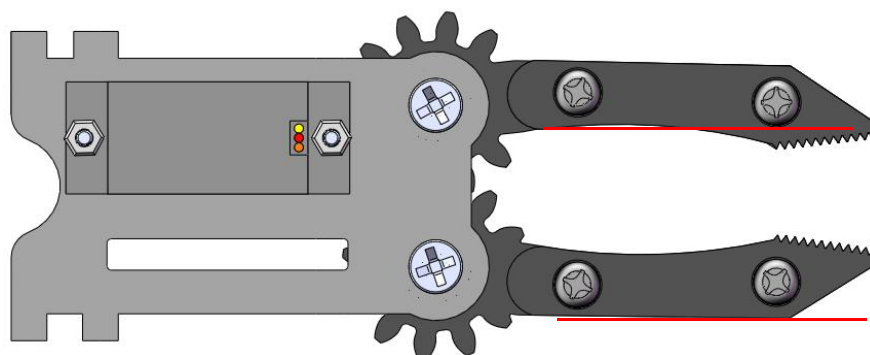
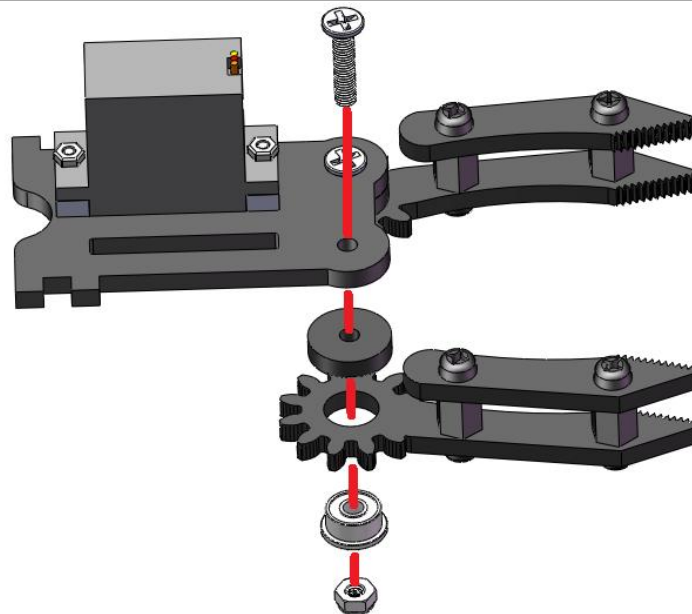
Step 13 Install the Left Gripper Plate of The Clamp

Step 10 Structure	Step 11 Structure	Bag No.⑭ Flange Bearing 1PCS
		
Acrylic 1PCS	Bag No.② M3X14MM Screw 1PCS	Bag No.⑦ M3 Lock Nut 1PCS
		



Step 14 Install the Right Gripper Plate of the Clamp

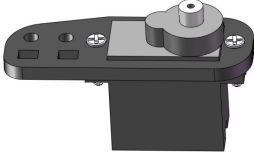
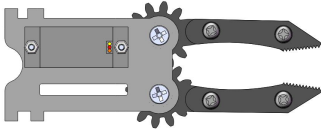
Step 13 Structure	Step 12 Structure	Bag No.⑭ Flange Bearing 1PCS
		
Acrylic 1PCS	Bag No.② M3X14MM Screw 1PCS	Bag No.⑦ M3 Lock Nut 1PCS
		

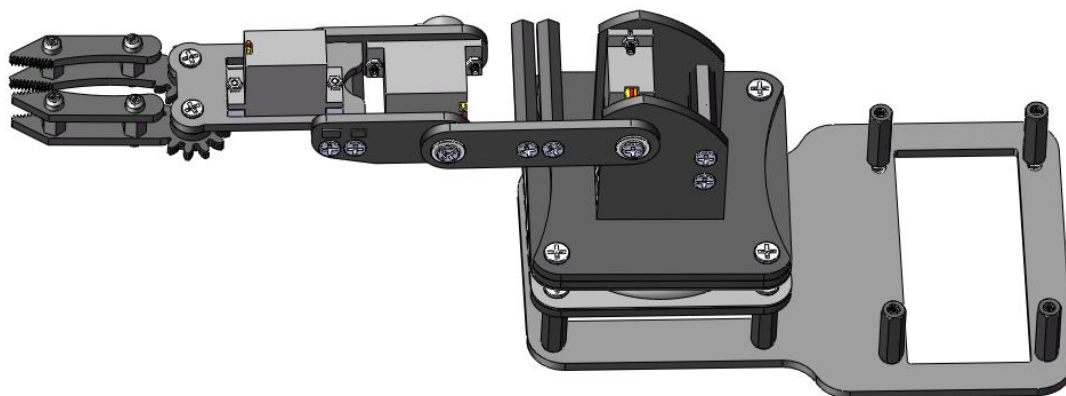
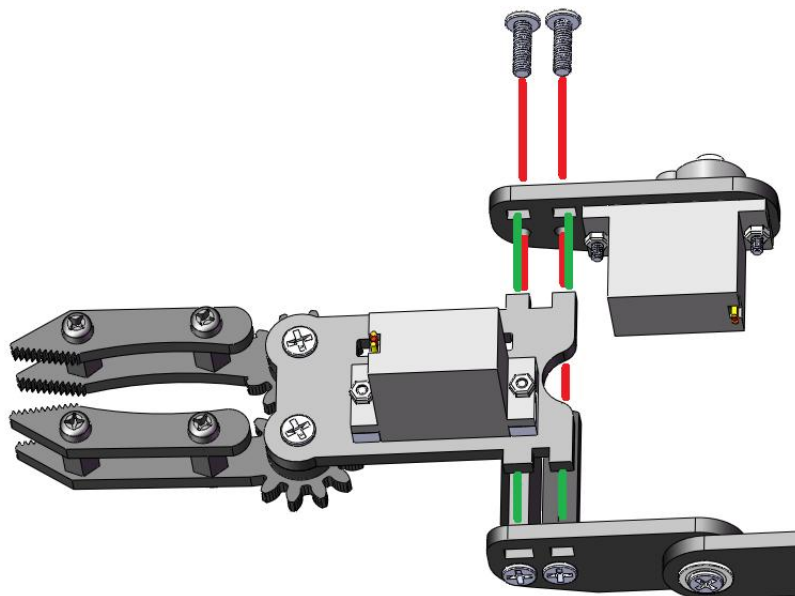


keep horizontal

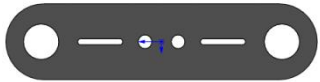
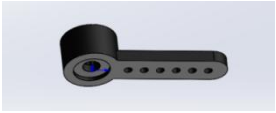
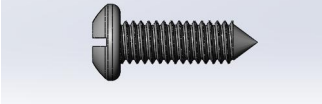
Note: The reference lines are intended for approximate alignment only. Due to normal manufacturing tolerances of the servo, slight angular deviations are expected. Exact alignment is not required and will not affect the normal operation of the robotic arm.

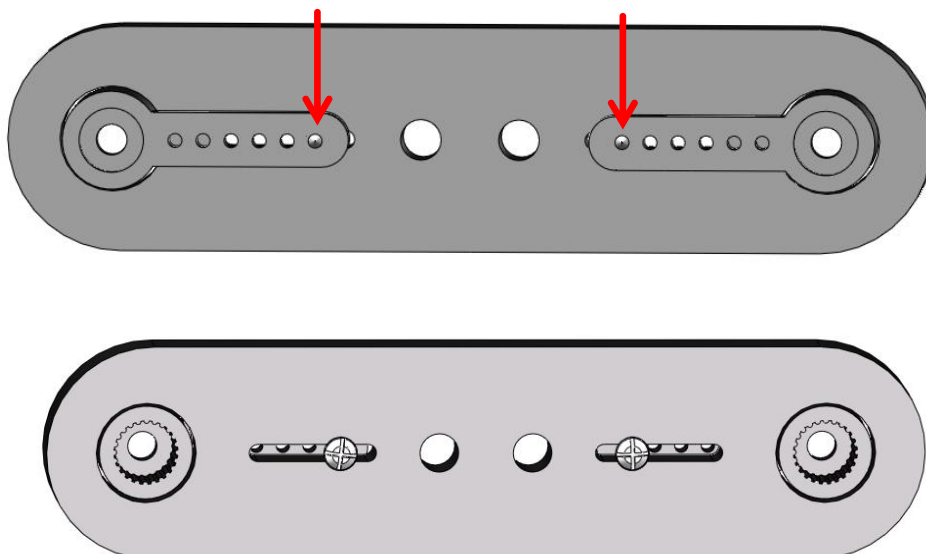
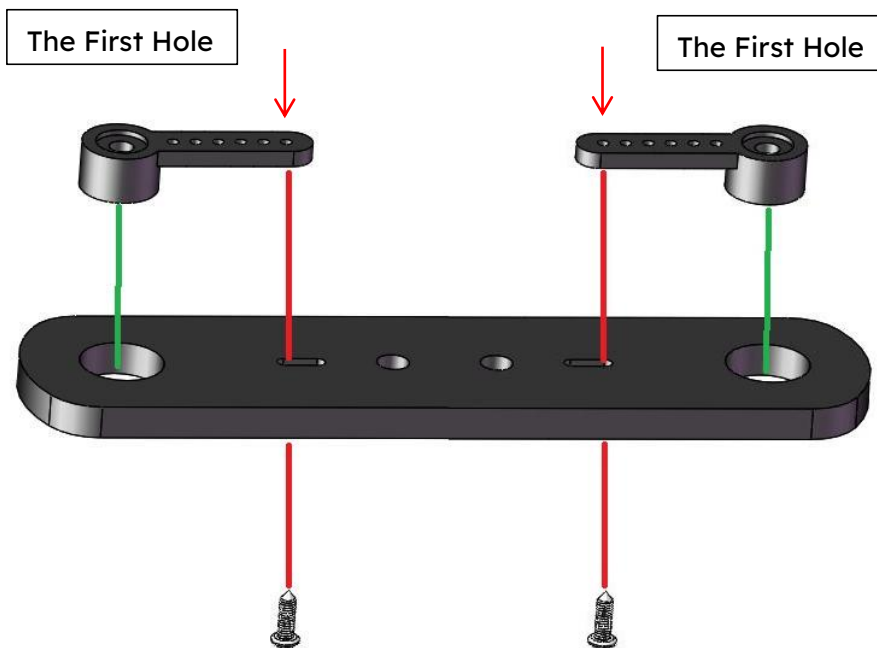
Step 15 Assemble the Clamp and Upper Arm

Step 8 Structure	Step 9 Structure
	
Step 14 Structure	Bag No.③ M3X10MM Screw 2PCS
	

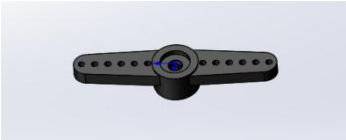
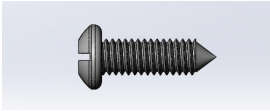


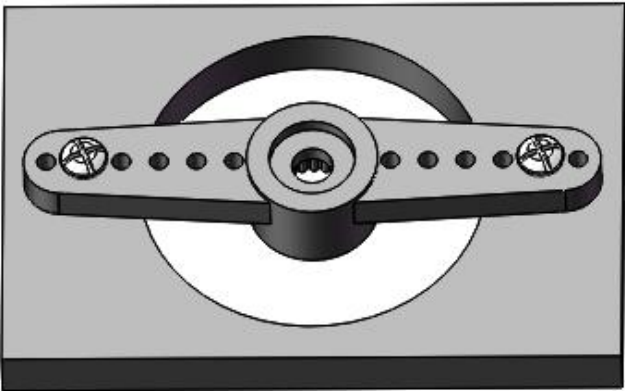
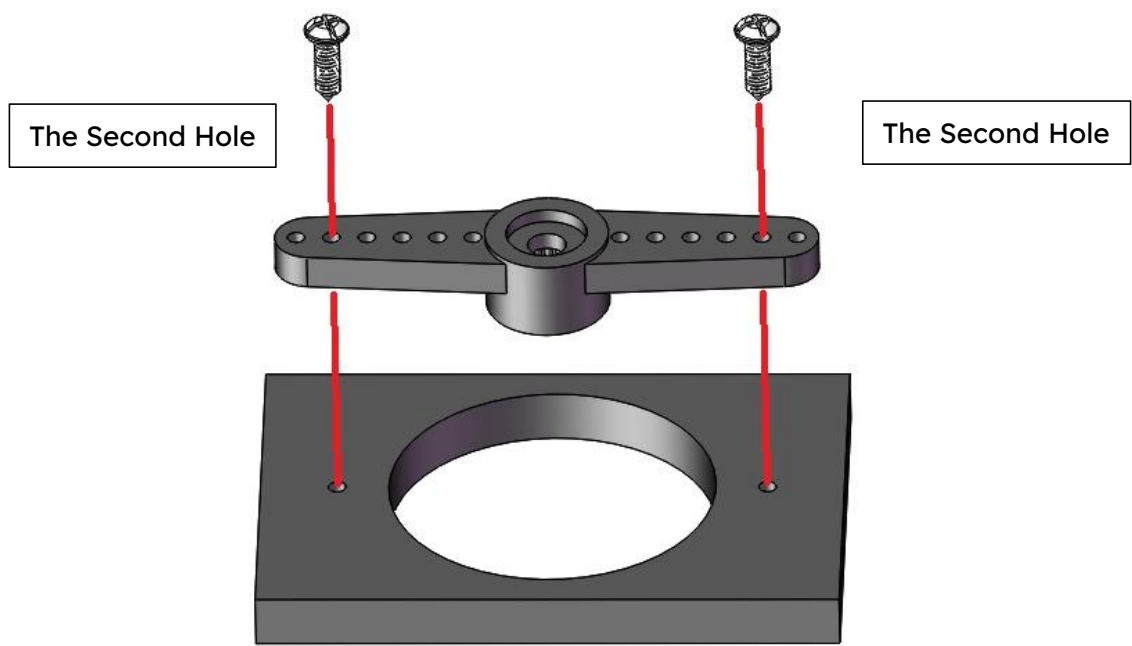
Step 16 Assemble the linkage between Servo B and Servo C

Acrylic Sheet 1PCS	Servo Horn 2PCS	Bag No.⑤ M1.4X5 Lock Screw 2PCS
		

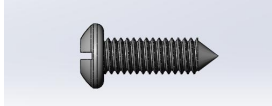


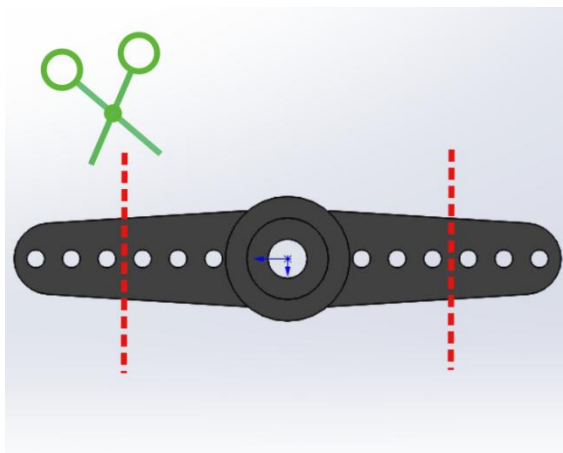
Step 17 Assemble the Mounting Structure for Servo A

Acrylic Sheet 1PCS	Servo Horn 1PCS	Bag No.⑤ M1.4X5 Lock Screw 2PCS
		

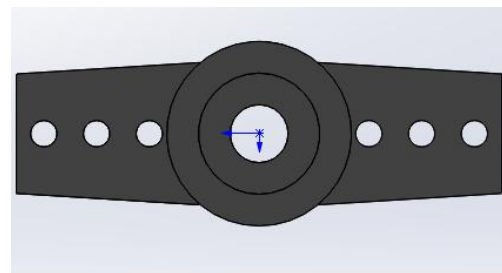


Step 18 Assemble the Mounting Structure for Servo D

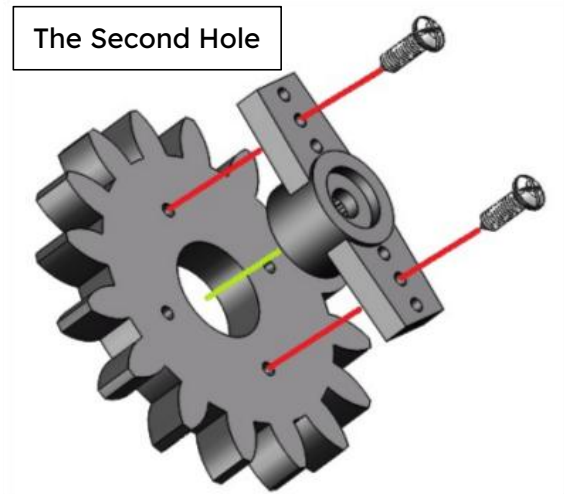
Acrylic Sheet 1PCS	Servo Horn 1PCS	Bag No.⑤ M1.4X5 Lock Screw 2PCS
		



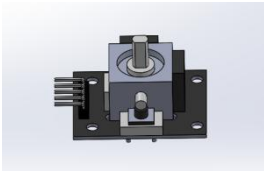
Pre-cut before factory shipment.

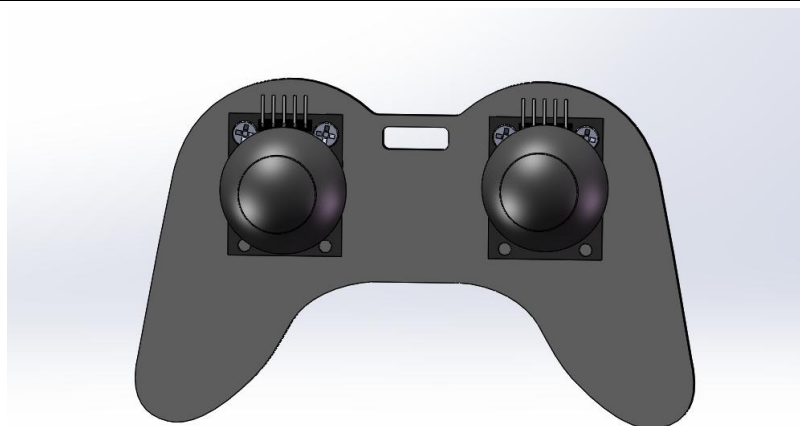
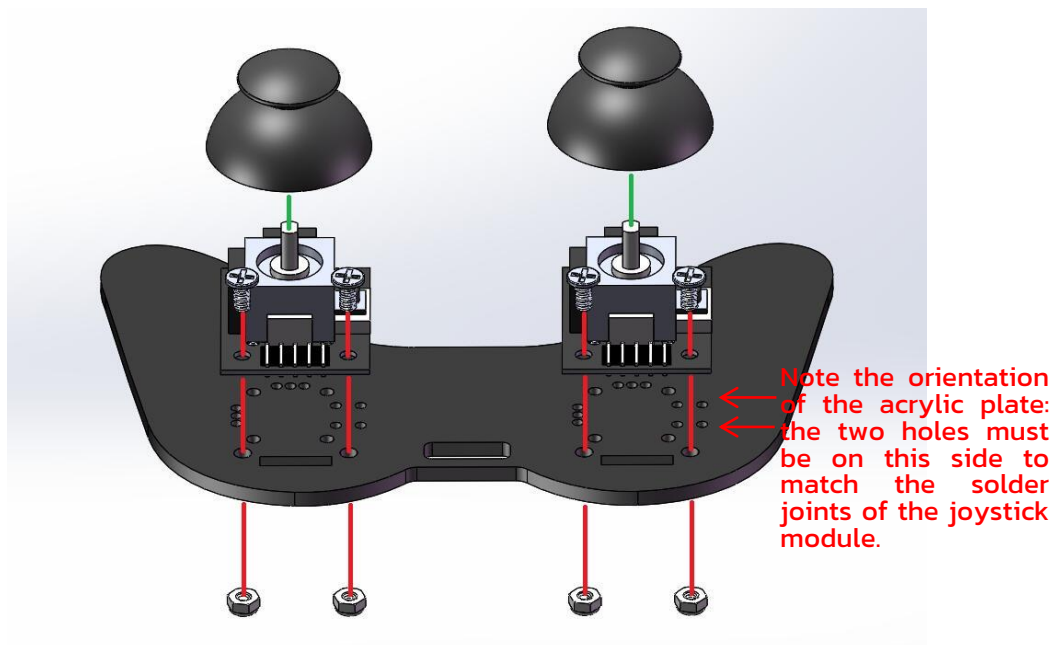


The Second Hole

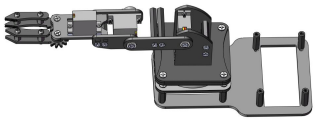
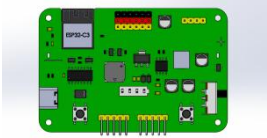
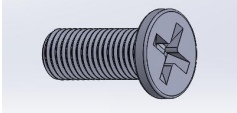


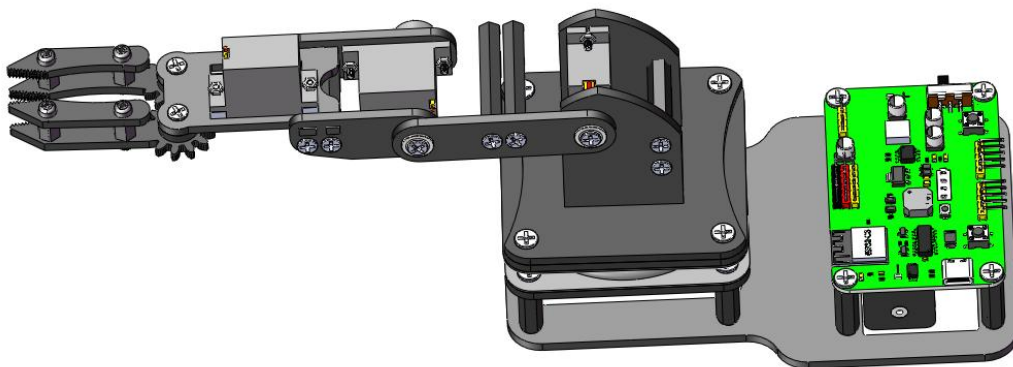
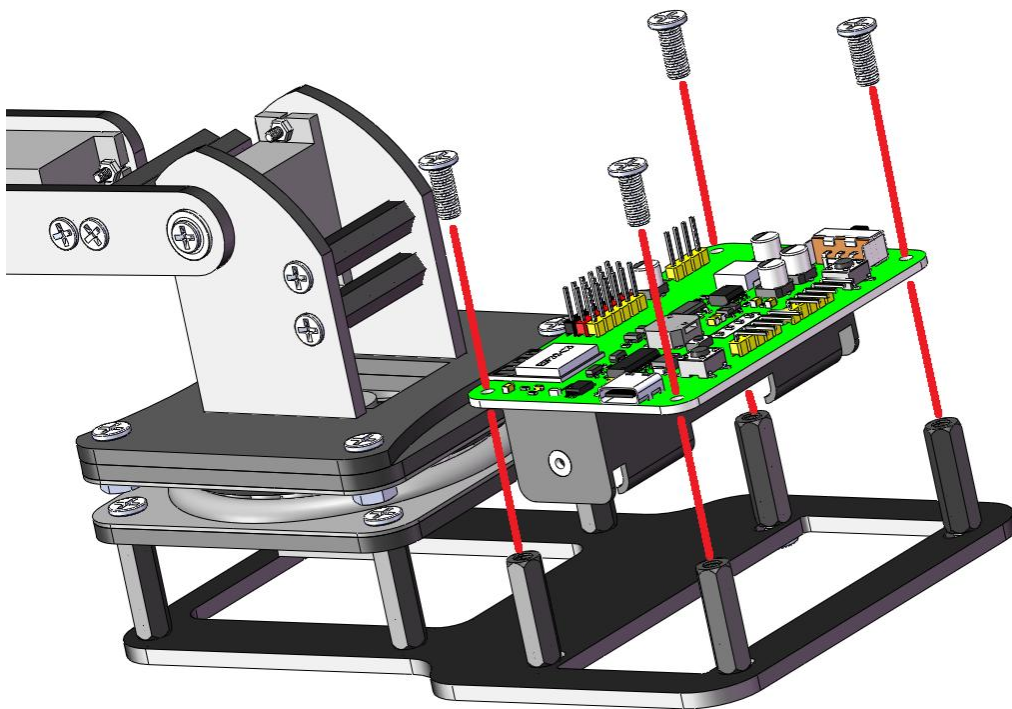
Step 19 Assemble the Joystick

Acrylic Sheet 1PCS		Joystick Module 2PCS	
			
Joystick Hat 2PCS	Bag No.③ M3X10MM Screw 4PCS	Bag No.⑦ M3 Lock Nut 4PCS	
			

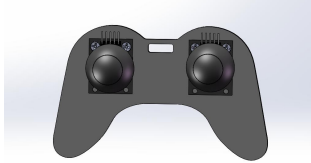
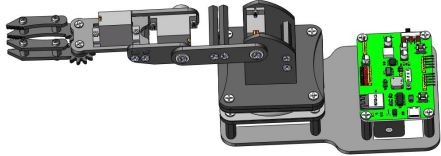


Step 20 Install the Esp32 Board

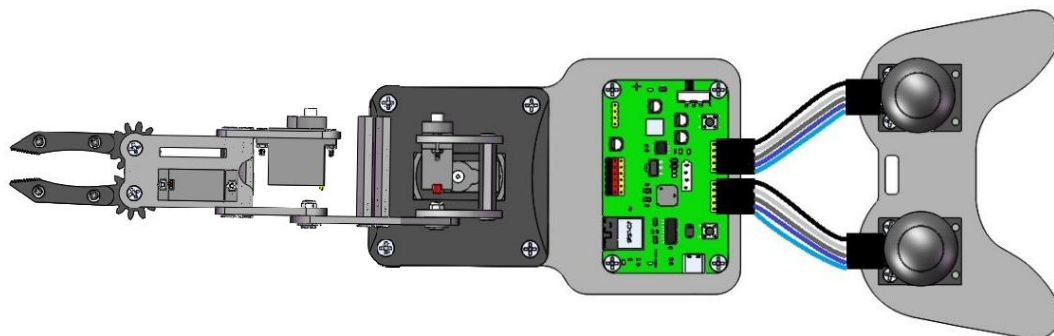
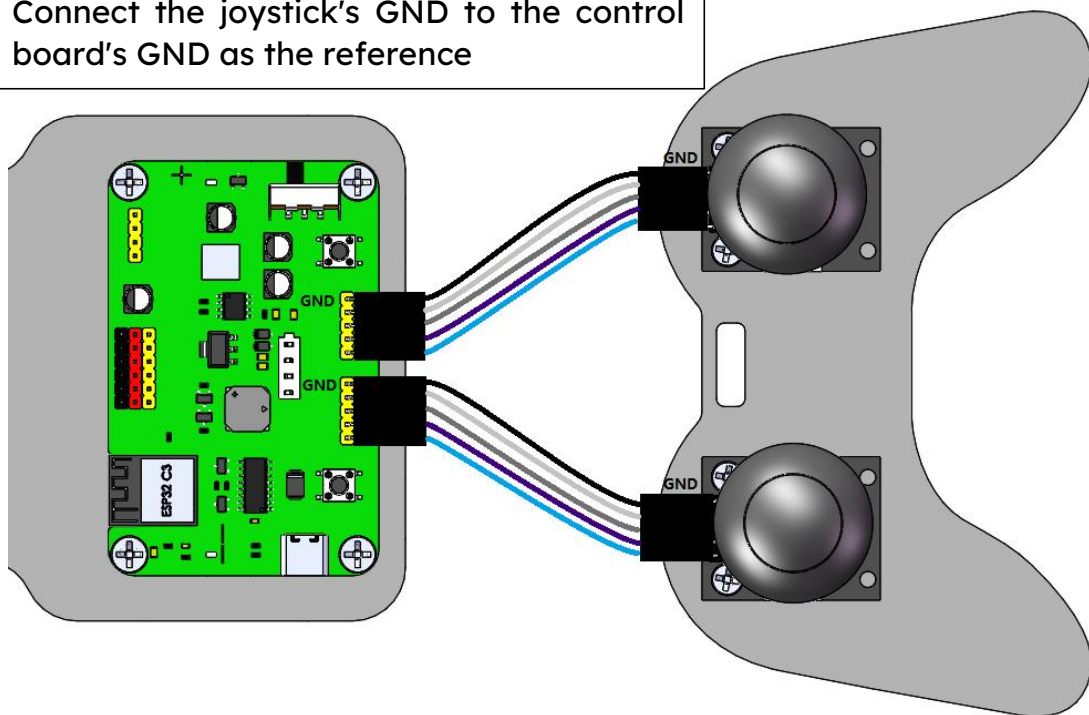
Step 15 Structure	ESP32 Control Board 1PCS	Bag No.① M4X10MM Screw 4PCS
		



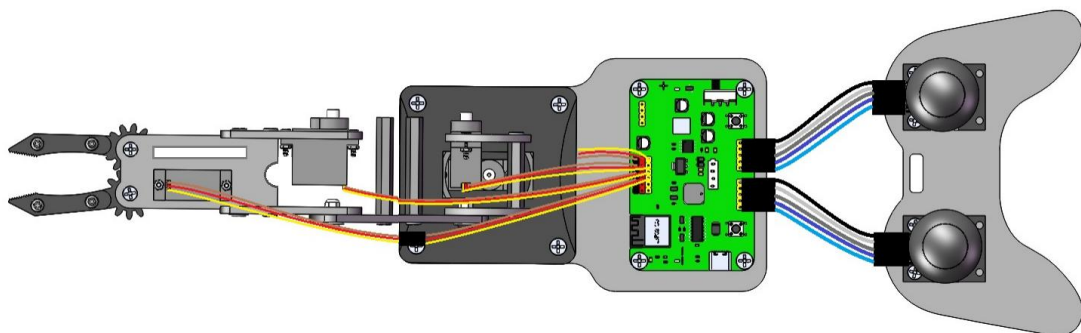
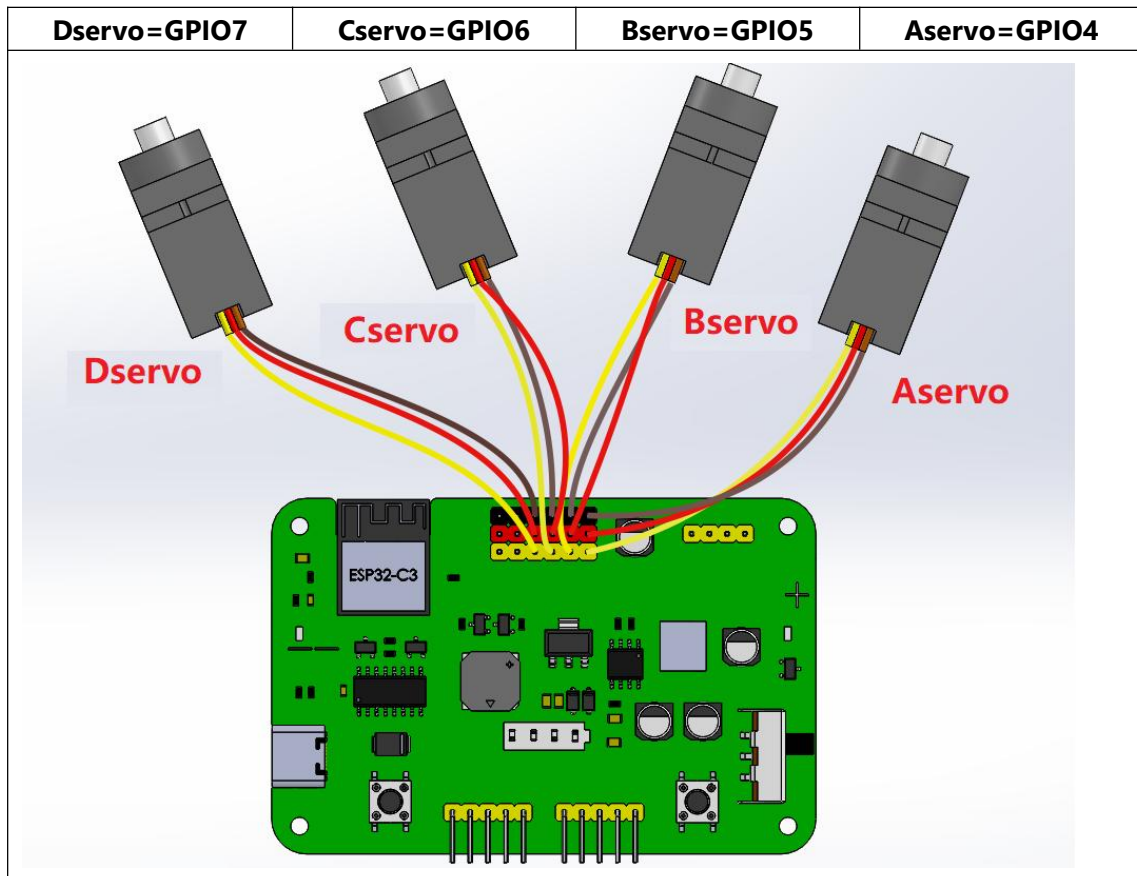
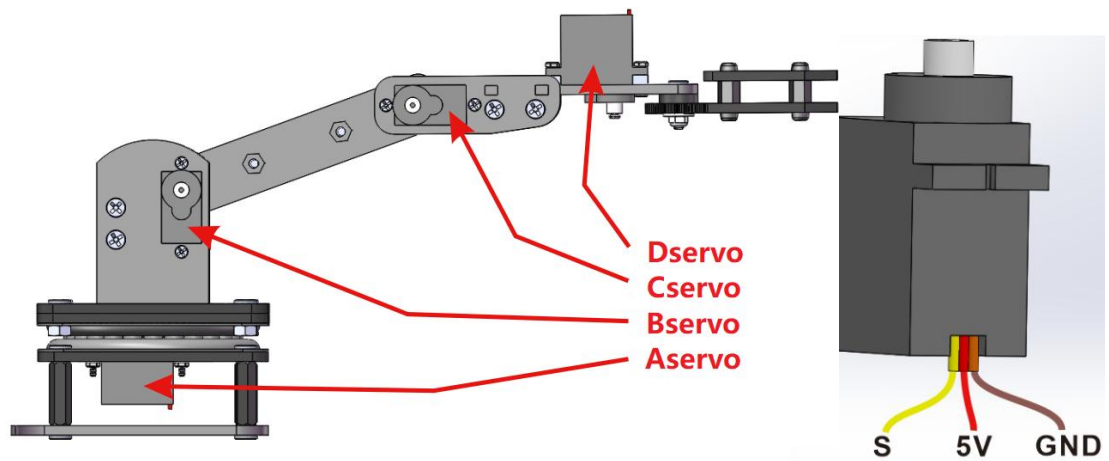
Step 21 Connect the Joystick

Part in Step 19	Part in Step 20
	
5P Dupont Wire 2PCS	
	

Connect the joystick's GND to the control board's GND as the reference

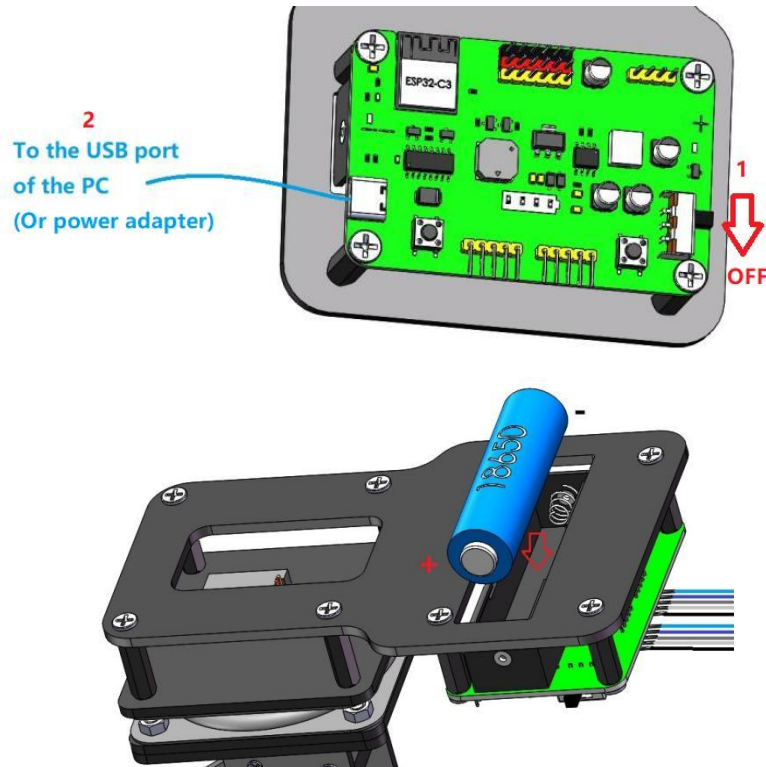


Step 22 Connect the Servos to the ESP32 Board



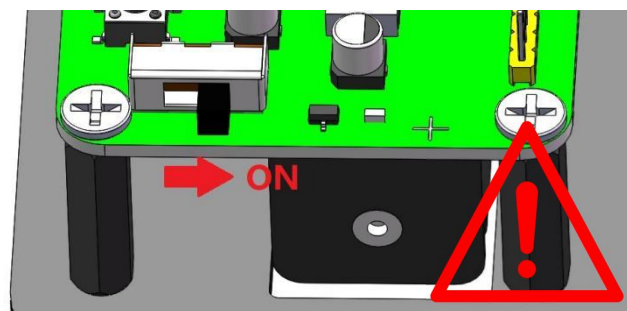
Step 23 Initialize the Servo Angle (Important!)

- 1) Turn the controller board switch OFF.
- 2) Install an 18650 lithium-ion battery (**not included**).
- 3) Before first use, fully charge the battery via USB cable.



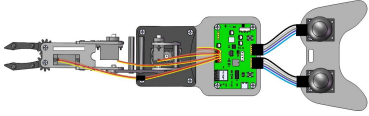
4) Important: Turn on the power switch!

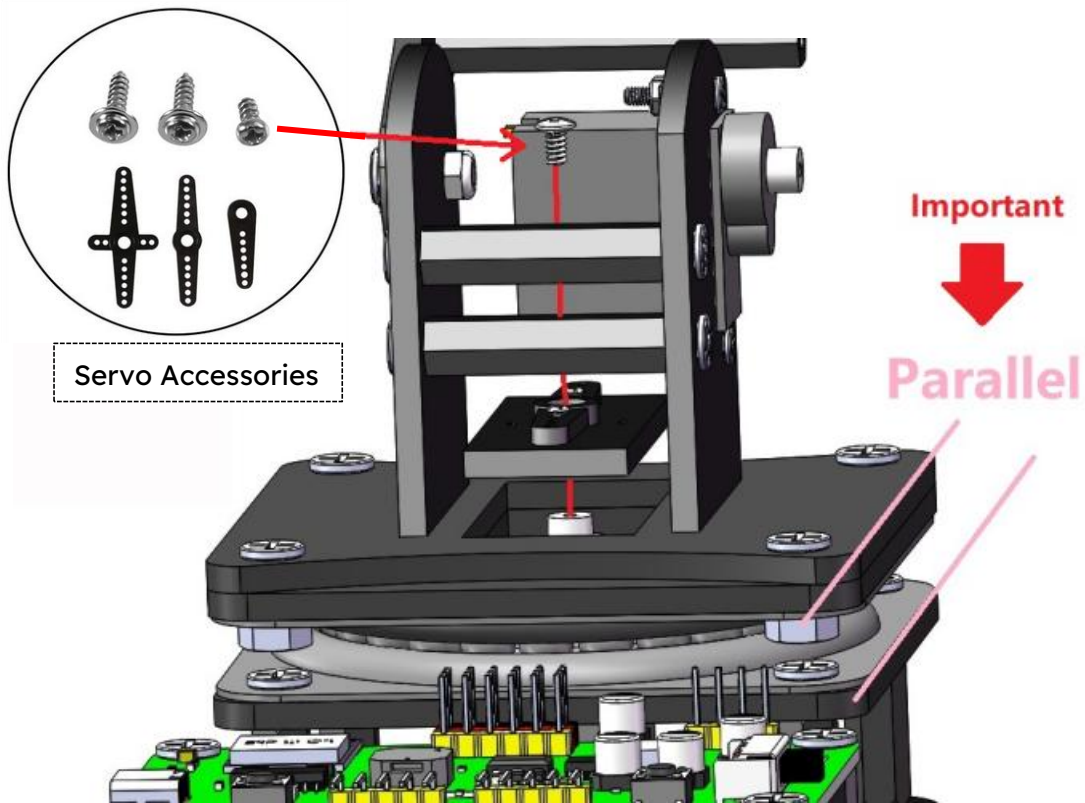
The control board comes pre-programmed with firmware. When powered on, the board will automatically position all servo motors at the **90-degree** default position.



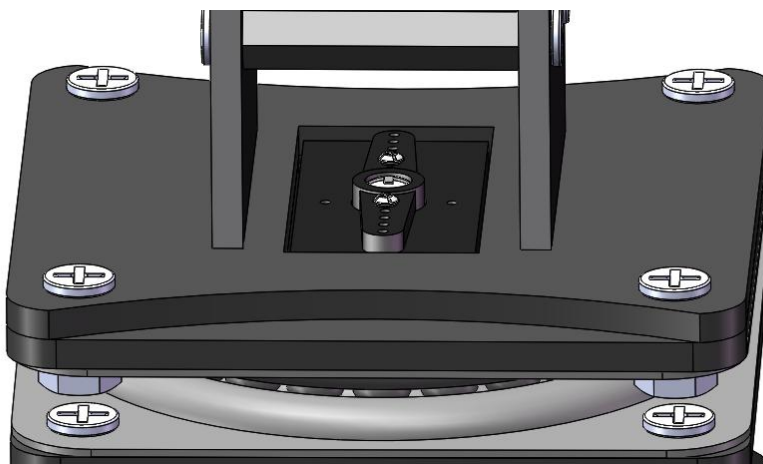
1. Make sure the battery is fully charged!
2. When powered on, the control board sends a 90° signal to all servos, causing them to rotate to the default position. You will hear the servo motor shaft rotating.
3. In the following installation steps, please keep the power on to prevent the servo angle from being changed!

Step 24 Fix the Robot Arm to Servo A of the Base

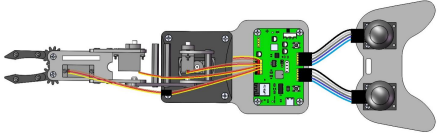
Step 23 Structure	Step 17 Structure	M2.5X4mm Screw 1PCS
		

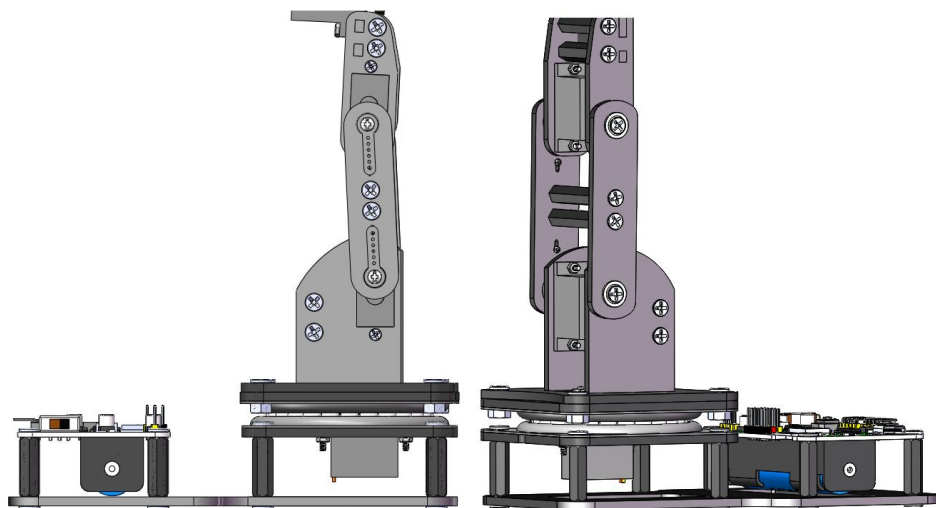
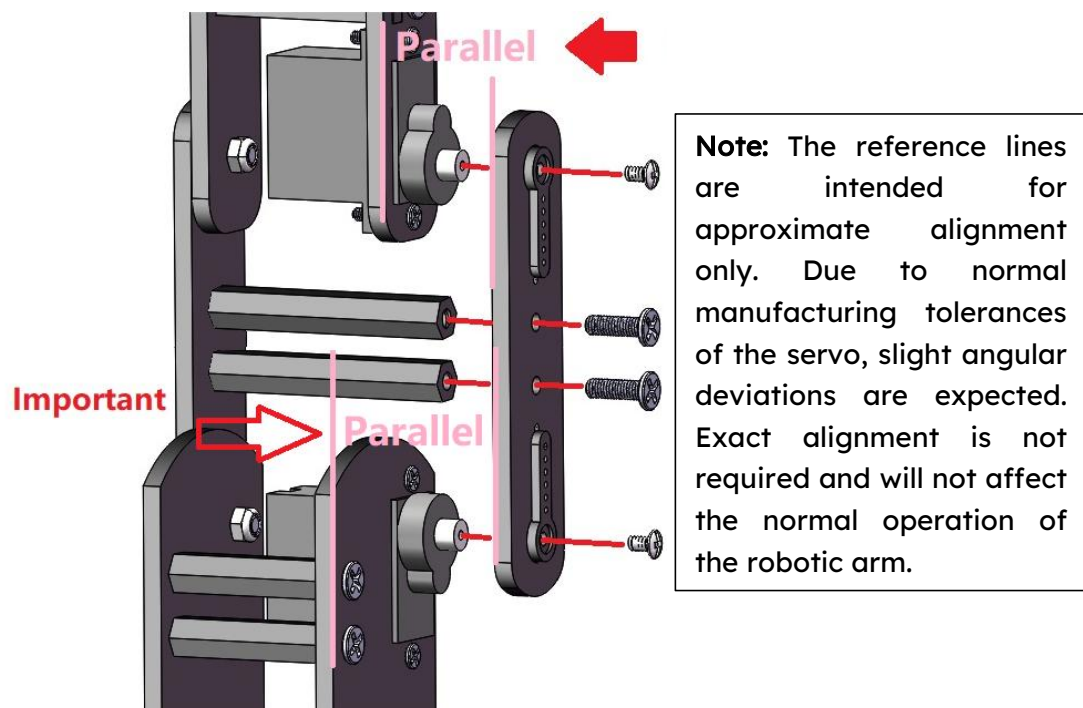


Note: The reference lines are intended for approximate alignment only. Due to normal manufacturing tolerances of the servo, slight angular deviations are expected. Exact alignment is not required and will not affect the normal operation of the robotic arm.

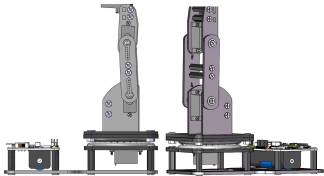


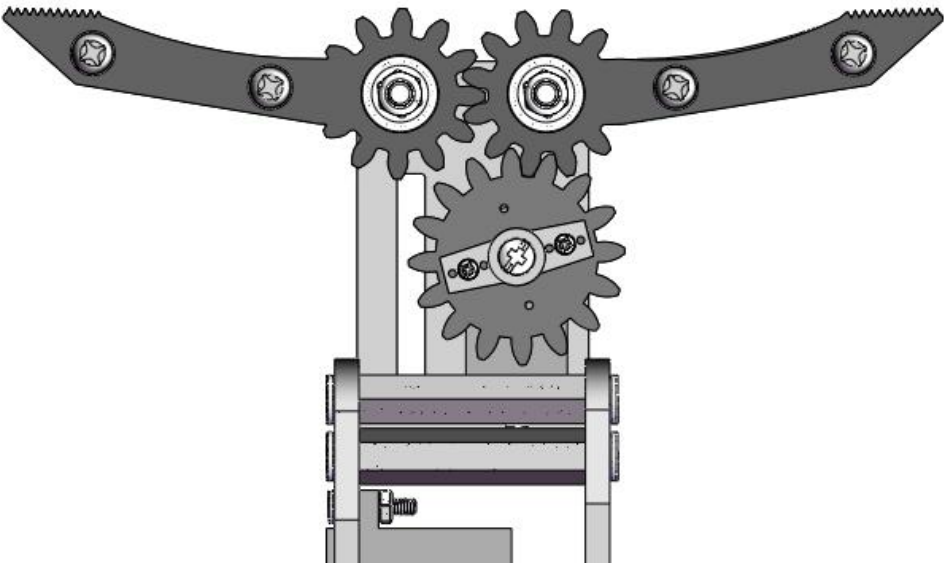
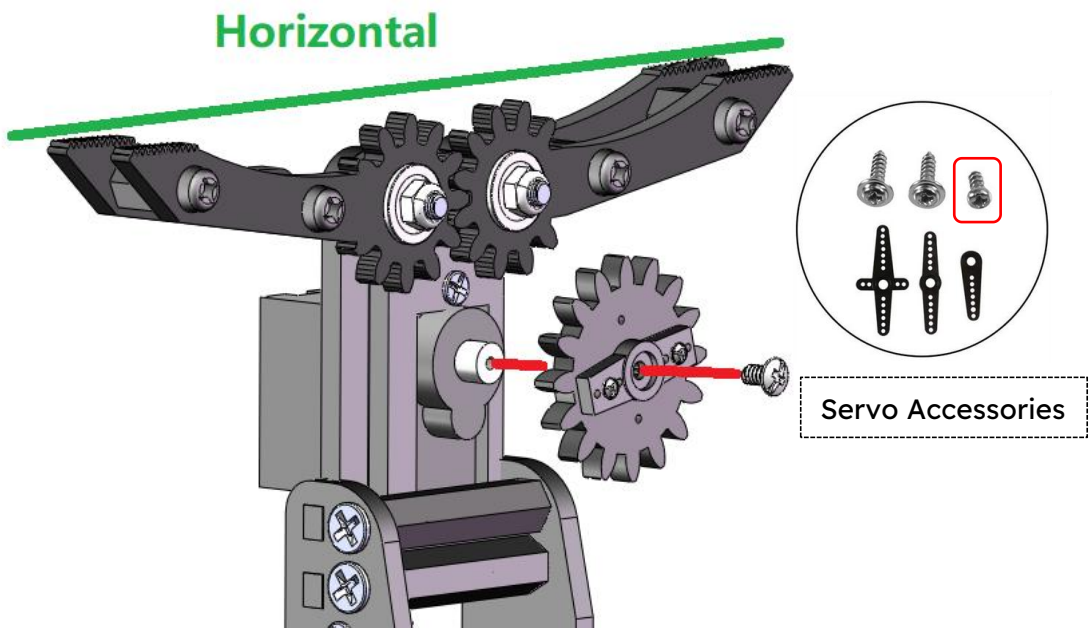
Step 25 Assemble the Middle Arm

Step 24 Structure	Step 16 Structure
	
Bag No.③ M3X10MM Screw 2PCS	M2.5X4mm Screw 2PCS
	

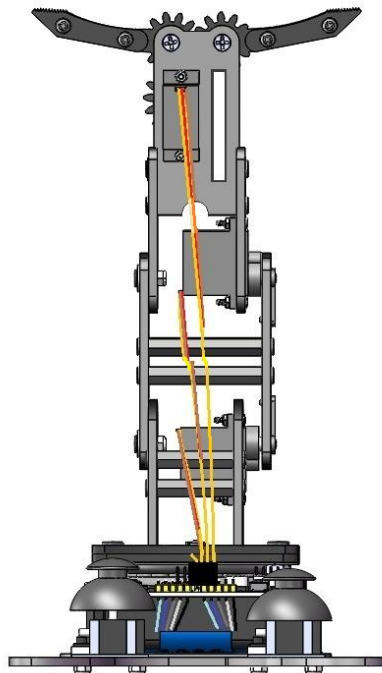
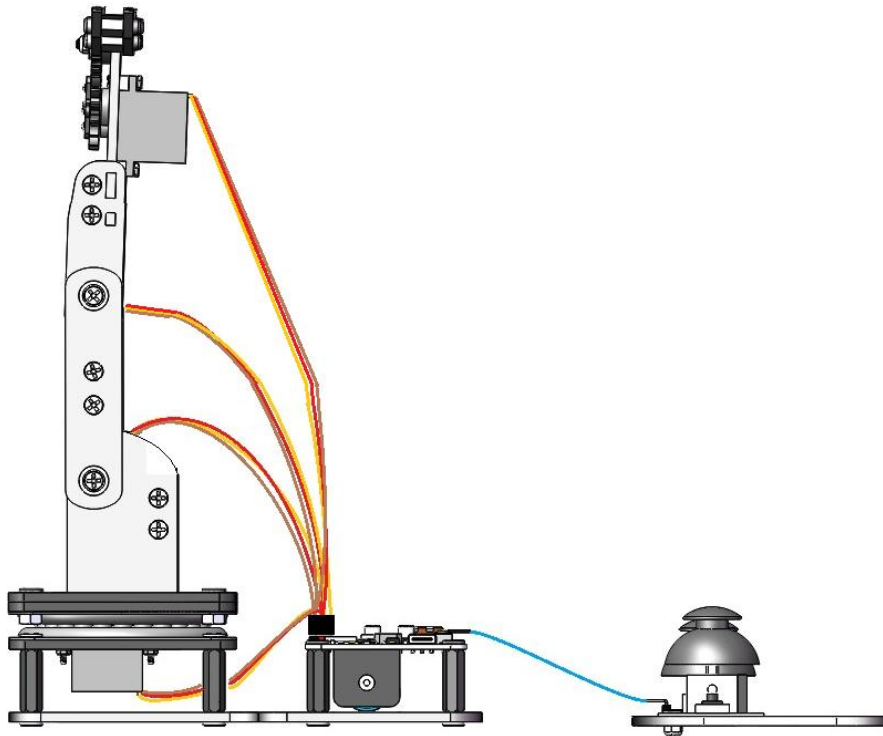


Step 26 Fix the Gripper Plates to Servo D

Step 25 Structure	Step 18 Structure	M2.5X4mm Screw 1PCS
		



Step 27 Assembly Complete



Using the eArm Robot

3.1 Robot Arm Safety and Operating Guidelines

To ensure optimal performance and maximize the service life of your ESP32 Robotic Arm, please read and follow these guidelines carefully.

1. Primary Operating Precautions

Do Not Overload: This arm is designed for lightweight tasks. The maximum recommended load for the gripper is 40 grams.

Maintain a Clear Workspace: Ensure the area around the robot arm is free of obstacles to allow unobstructed movement of all joints.

Avoid Forced Movement: Never manually force or twist any joint while the arm is powered on, as this may damage internal gears and servo motors.

2. Critical Gripper Maintenance (Servo Protection)

Continuously gripping an object under load places significant stress on the servo motor (particularly the D-axis driving the gripper), causing it to generate heat. Therefore, we strongly recommend limiting continuous load-bearing gripping to no more than 40 seconds. After completing a gripping operation, release the object immediately by opening the gripper to allow the motor to cool down and rest. Exceeding this duration is the primary cause of motor overheating and burnout.

Auto-Protection Function: Our latest optimized firmware includes a protective feature. If the gripper servo does not receive a new command within 40 seconds, it will automatically disengage (cease PWM signal transmission) to prevent overheating.

Best Practice: Actively control the gripper. When not manipulating objects, open the gripper or send commands to maintain it in a relaxed state.

3. Routine Maintenance

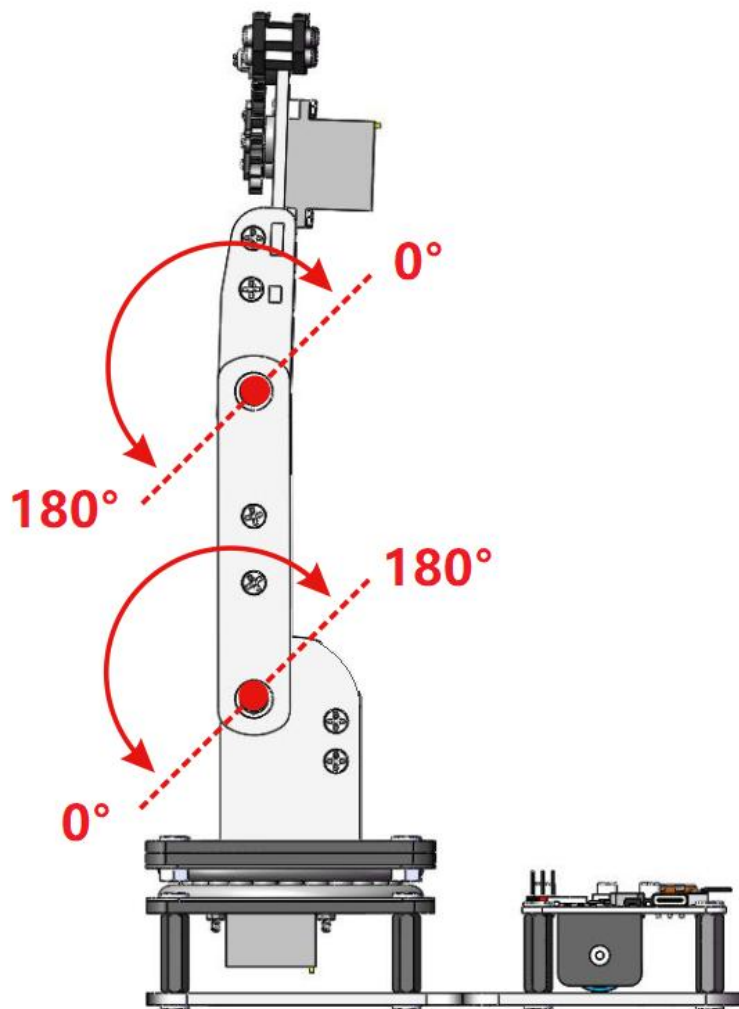
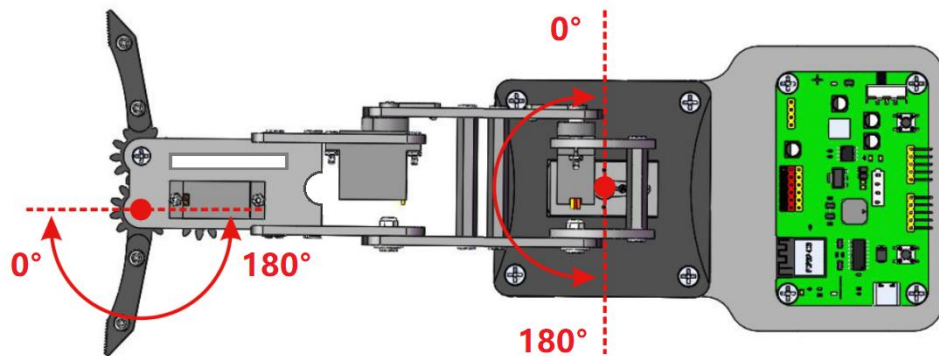
Power off the unit when not in use and remove the battery.

Periodically check and tighten all screws and connections.

Operate the robot arm on a stable, flat surface.

Following these instructions will help ensure a reliable and durable robotic experience.

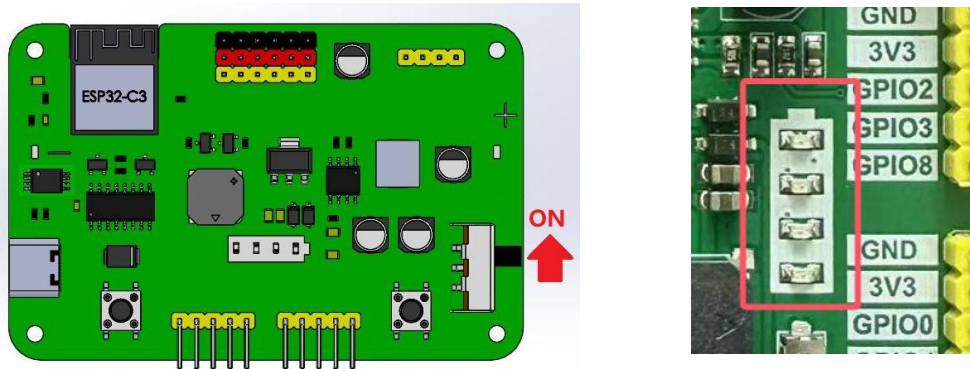
3.2 Degrees of Freedom



3.3 Control the eArm Robot Arm with a Joystick

Turn the eArm robot's power switch to the **ON** position.

- 1: Make sure the 18650 lithium battery is installed.
- 2: The battery indicator shows 3 bars or more.

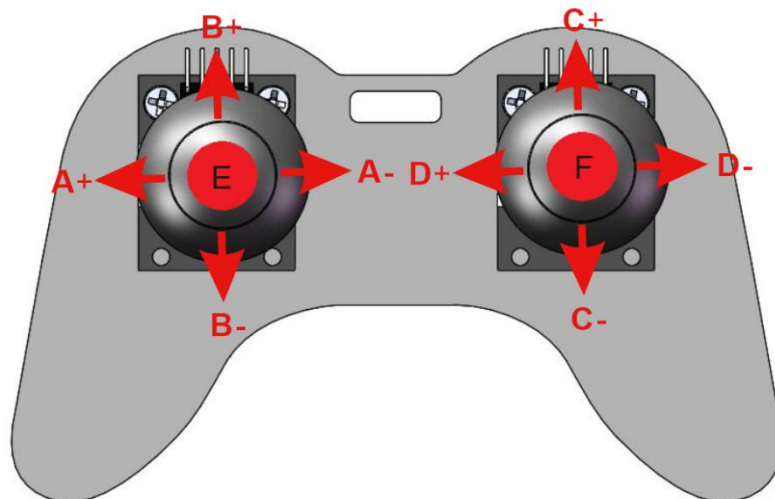


After powering on the eArm robot, the default control mode is **Joystick Mode**. To switch between control modes, press the "**F**" button on the right joystick.

Available modes include:

- ✓ ■ Joystick Mode (default)
- ✓ ■ Web App Mode

When you press "**F**", the buzzer will beep once, indicating the switch to **Web App Mode**. In this mode, joystick controls will be **disabled**.

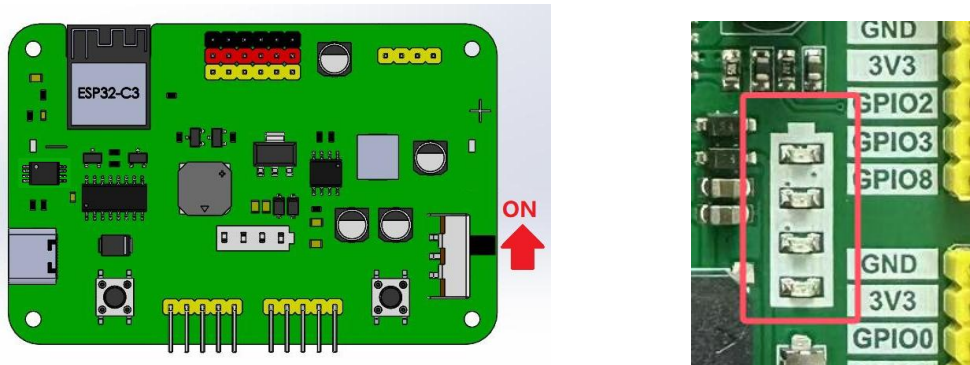


- | | |
|--|---|
| A+ : Rotate arm left (Servo A) | A- : Rotate arm right (Servo A) |
| B- : Raise rear arm (Servo B) | B+ : Lower rear arm (Servo B) |
| C- : Raise the forearm (Servo C) | C+ : Lower the forearm (Servo C) |
| D+ : Open the gripper (Servo D) | D- : Close the gripper (Servo D) |
| F : Toggle between Joystick and Web App control modes | |
| E : Enable/disable the buzzer | |

3.4 Control the eArm Robot Arm with Web App

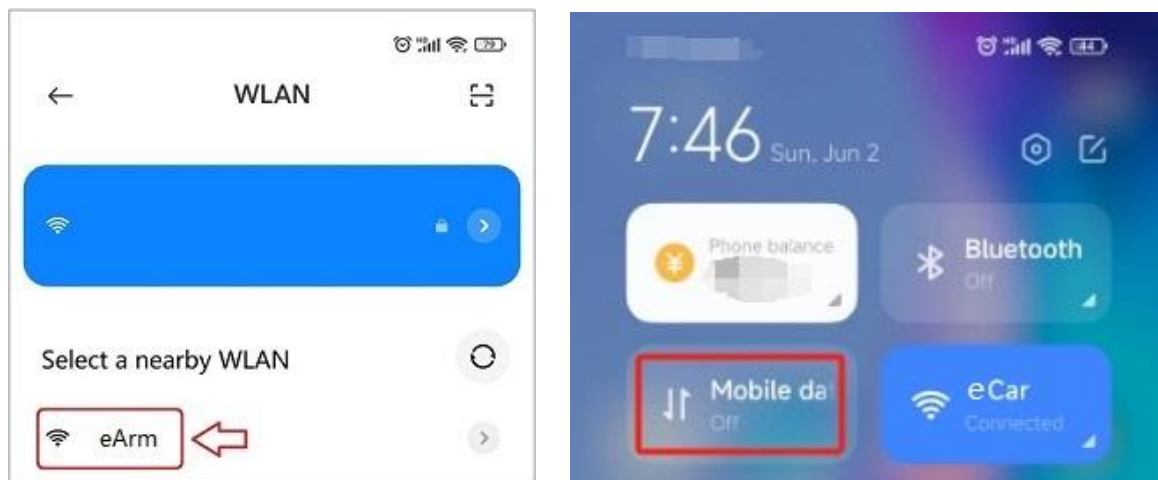
Turn the eArm robot's power switch to the **ON** position.

- 1: Make sure the 18650 lithium battery is installed.
- 2: The battery indicator shows 3 bars or more.

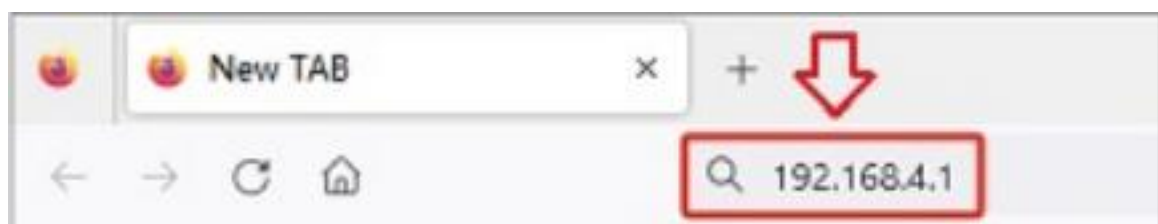


Turn on the phone's Wi-Fi and connect to the network named '**eArm**'

- : Once connected to the Wi-Fi, you may see a '**Network connection failed**' message; **just disregard it**.
- : Step 1: **Turn off mobile data** in your phone settings
- : Step 2: This ensures stable connection to the Web App



- : Step 3: Open the web browser on your phone
- : Step 4: Type **192.168.4.1** into the address bar
- : Step 5: Press Enter to connect



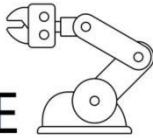
After successful connection, the control interface will appear in your browser - you can now operate the eArm robot arm!

5:39

Q eArm Control

⋮

eArm
SIYEENOVE



P: 100%



A+

A-

B+

B-

C+

C-

D+

D-

<

☆

➦

Button	Mechanical Movement
A+	Robot arm turns left
A-	Robot arm turns right
B-	Rear arm raises
B+	Rear arm lowers
C-	Front arm raises
C+	Front arm lowers
D+	Gripper opens
D-	Gripper closes

Press the F button again to switch back to Joystick Mode